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Ontology-enhanced RAG for a personalised and sustainable food advisory system ☆

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ABSTRACT

Supporting consumers in making autonomous food choices that are sustainable and nutritionally complete is an increasingly complex task that must take into account several needs to foster eating habits of health-conscious consumers, while reducing food waste and environmental impact. While Generative AI and Large Language Models (LLMs) show promising results in this domain due to their natural language processing capabilities, they suffer from critical limitations, including hallucinations, knowledge gaps, and limited ability to handle factual information. To mitigate such limitations, Retrieval-Augmented Generation (RAG), which retrieves relevant information from external sources to enhance the capabilities of LLMs, has shown effectiveness in many domains. However, existing RAG approaches typically operate on unstructured text that lacks sophisticated symbolic representations of complex domain knowledge. This work proposes an ontology-enhanced conversational food advisory system that integrates a modular ontology, named FoCOSA (Food Consumer-Oriented Sustainability-Aware), within several key tasks of a RAG-based system, enhancing LLM reasoning with domain knowledge, while simultaneously improving the interpretation of user requests, thus improving retrieval effectiveness and interaction fluidity. Experimental evaluations demonstrate the efficacy of the approach, and the study concludes with guidelines for selecting appropriate settings for food recommendation scenarios considering the complexity of natural language queries and other contextual factors.

1. Introduction

Supporting consumers in making autonomous food choices that are sustainable and nutritionally complete is an increasingly complex task that must take into account several needs of health-conscious consumers [1]. In fact, consumers may have overlapping preferences such as dietary restrictions due to intolerances (e.g., lactose intolerance), cultural aspects (e.g., related to the customer's religion) and specific diseases that may be affected by food habits (e.g., diabetes condition), as well as nutritional needs (e.g., high-protein diets), and sustainability goals [2,3]. Today's technology-driven society increasingly relies on artificial intelligence (AI) to deliver personalised food suggestions [4]. These systems may provide overly generic recommendations, such as suggesting lactose-free products, without considering the user's need for sufficient protein intake, or overlook sustainability goals, such as products with reduced environmental impact and minimised food waste. Conversely, some systems over-personalise advice based on historical user data, creating recommendation bubbles that limit exposure to new and healthier options [5].

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In this context, providing suggestions to perform food choices requires the integrated usage of information coming from several different sources and tools that are able to support food advising tasks in a conversational manner. The former issue has been largely addressed through formal representation of knowledge in terms of concepts and relationships, with the possibility of enhancing domain models with semantics and reasoning facilities, using tools like ontologies and knowledge graphs. On the other hand, with the advent of Large Language Models (LLMs) such as ChatGPT, Claude, and Gemini [6], researchers have started to explore how these models can address the latter issue, that is, providing food advice in a conversational manner by relying on the Natural Language Processing capabilities of LLMs [7,8]. Despite their potential, these models have known limitations, including hallucinations, knowledge gaps, inadequate handling of factual information, and outdated knowledge. To address these challenges, Retrieval-Augmented Generation (RAG) enhances LLMs by incorporating external knowledge sources into the generation process, thereby improving factual accuracy and reliability [9]. However, RAG-based approaches normally operate on unstructured textual data and do not leverage symbolic representations for deeper domain understanding. This limitation makes it difficult for RAG to comprehend and summarise semantic concepts across large datasets [10].

To address these challenges, researchers have explored the joint use of LLM and ontologies or knowledge graphs. Recent examples include ontology-driven interactions with LLMs [11,12] and ontology-based refinement of knowledge graph queries [13]. On the one hand, traditional ontology-based reasoning provides structured, deterministic inferences, but struggles with handling natural language ambiguity, and multi-faceted queries. Large Language Models (LLMs), on the other hand, enable flexible interpretation of user intent, but often lack factual consistency and structured domain knowledge. An ontology-only approach requires strictly structured queries (e.g., SPARQL), making it difficult for non-expert users to interact naturally. Additionally, ontology-based reasoning alone struggles when dealing with missing or ambiguous information, whereas LLMs can generalise beyond predefined rules. While these techniques have shown promising results in general knowledge integration, there remains a gap in comprehensively combining nutritional, sustainability, and personal preference factors within a single system. Moreover, the potential of combining advanced LLMs, offering a conversational interaction modality, with domain-specific ontologies for providing food advice is still largely unexplored.

This work proposes an ontology-enhanced conversational food advisory system, that integrates a modular ontology within a RAG framework, enhancing LLMs' ability to comprehend context and provide more relevant and reliable food-related information. With respect to the state-of-the-art ontology-enhanced LLMs, our approach employs the ontology throughout the RAG pipeline: the ontology provides structured domain knowledge to enhance LLM reasoning and ensures semantic consistency, while the LLM leverages its natural language understanding to interpret complex user queries and enable fluid conversational interaction. Specifically, our method employs the ontology to support three key phases of the RAG process: semantic description generation to improve indexing, query rewriting to optimise information retrieval, and response generation to ensure alignment with ontological concepts and their mutual semantic relationships.

The central role in this approach is played by the FoCOSA (Food Consumer-Oriented Sustainability-Aware), a modular ontology, that integrates aspects such as health conditions, physical activities, job-related lifestyle constraints, and food production sustainability concepts (e.g., environmental, social, economic and waste sustainability). This comprehensive knowledge representation enables the system to help users make informed food-related decisions that aim at improving their nutritional habits while respecting sustainability goals. The modular design of the ontology is crucial to handle its inherent complexity and to ensure its evolution over time. By breaking the ontology into smaller, self-contained modules, we enable flexible management and easier maintenance.

Through an experimental evaluation, we analyse the impact of ontology integration at different phases of the RAG pipeline on improving response accuracy, reducing hallucinations, and providing contextually aware and sustainable food insights. While existing RAG evaluation frameworks, such as RAGAS [14], provide useful metrics, they are not fully suited to the assessment of complex, open-ended generative outputs in advisory settings, such as the one addressed in this work. Some metrics rely on ground-truth annotations requiring substantial expert involvement and resources not yet available at the current stage of our research, while others produce binary judgements offering limited interpretability for comparative analysis across system configurations. For these reasons, we adopt an LLM-as-a-Judge strategy [15] which does not demand the availability of ground-truth information (see Section 7.1.1). Operatively, we employ an LLM-as-a-Judge approach using a multi-level ordinal scale (namely, a 0–5 Likert scale), which allows us to produce graded, interpretable evaluations of LLM-generated advice. This approach has been widely adopted in previous studies and recommended as a practical standard for comparing model performance on human-centred quality dimensions [16]. Furthermore, while RAGAS provides standard evaluation criteria that do not require ground-truth information, these criteria remain largely general-purpose and are not explicitly tailored to assess the effectiveness of domain-specific personalised advisory scenarios. We therefore introduce specific criteria (e.g., *Actionability* and *Data Utilisation Effectiveness*), designed to capture the unique characteristics of food advisory systems. The following section grounds these technical innovations in concrete application scenarios.

The main contributions of this work are:

- an ontology-enhanced RAG framework for food advisory, supporting users in making informed food-related decisions based on health conditions, physical activities, and food production sustainability;
- the integration of the FoCOSA modular ontology in different RAG stages to improve information retrieval and enhance domain-specific reasoning;
- an empirical evaluation demonstrating the impact of ontology-enhanced RAG in improving response accuracy and reducing hallucinations.

The remainder of this paper is structured as follows: Section 2 presents motivating examples, outlines the three-phase methodological framework and provides an architectural overview of the system; Section 3 discusses the core foundational approaches relevant to our proposal, while Section 4 illustrates the most recent work on the combined use of food ontologies and LLMs; Section 5 introduces the modular ontology FoCOSA and its three core modules; Section 6 describes the architecture and the implementation details of the proposed system; Section 7 presents the experimental evaluation and results; finally, in Section 8 we provide our conclusions and delineate possible future research directions.

2. System design: Motivation, methodology, and architecture

This section establishes the foundation for the technical contributions in concrete user scenarios (Section 2.1), then outlines the three-phase methodological framework guiding system development (Section 2.2) and provides a high-level architectural overview (Section 2.3).

2.1. Motivating example

We consider two representative consumers whose dietary needs illustrate the multifaceted nature of food-related decision-making. First, Sara is a 35-year-old consumer who is following a vegetarian and gluten-free diet, motivated by personal choices and a health condition, respectively. In addition, Sara is deeply committed to environmental sustainability and seeks to make informed and responsible decisions. She actively seeks food options that minimise carbon and water footprint, favouring locally sourced and organic products whenever possible. Sara, as a consumer, may ask for broad and multifaceted questions addressing sustainability, nutrition, and her shopping behaviour. A representative query from Sara may be the following one: *"Do my food purchases align with my sustainability goals, gluten-free and vegetarian eating preferences, and how can I minimise waste?"*. This type of question requires reasoning across several ontology modules, integrating knowledge on consumer's preferences, environmental impact of food production, and food product composition. This scenario highlights the need for interpretable, actionable recommendations to balance complex dietary needs with strong sustainability preferences. Second, Elizabeth is a 42-year-old consumer affected by diabetes and heart disease. Her primary goal is to receive reliable and medically appropriate food suggestions aligned with her health conditions and dietary restrictions. Her needs are to maintain stable blood sugar levels and cardiovascular health. She expects the system to provide safe and personalised suggestions based on her health profile and nutritional needs. For example, a representative Elisabeth's query might be the following one: *"How does salmon impact the health of consumers with cardiovascular conditions and diabetes?"*. This scenario highlights the relevance of healthcare grounded dietary suggestions.

Together, these examples highlight that consumers face multifaceted and often overlapping factors when making food choices, such as dietary requirements, health conditions, and sustainability concerns. These factors translate into concrete research challenges for a food advisory system:

- providing nutritional and shopping suggestions tailored to individual dietary requirements (*personalised nutrition guidance*);
- offering clear insights into sustainability metrics and preferences, enabling conscious choices (*sustainability transparency*);
- enabling empowerment of consumers, who may benefit from coherent integration of ontology modules (*consumer empowerment and awareness*).

2.2. Methodology overview

Problem statement. The main research problem addressed in this work concerns how to integrate heterogeneous food, nutrition, and sustainability data with formal domain knowledge and LLM-based reasoning to provide personalised and semantically consistent recommendations. Bridging these paradigms requires not only a technical solution but a methodological framework capable of capturing the interplay between explicit, ontology-based reasoning and contextual, LLM-driven inference required in this work. To address this, we propose a methodology to establish a unified research and engineering framework that connects conceptual modelling, data processing, and generative inference under shared criteria of semantic integrity.

Phases of the methodology. The methodology is organised into three interconnected phases: Conceptual, Engineering, and Evaluation—each comprising well-defined steps with specific inputs, processes, and outputs.

1. Conceptual phase

The Conceptual phase defines the theoretical and semantic foundation of the system. It defines the formal ontology, its domain coverage, and the conceptual artefacts that structure the hybrid reasoning pipeline. Its main activity is ontology and domain modelling, which transforms expert knowledge about food, nutrition, and sustainability into a formal representation. This phase is structured into two main steps: i) *Knowledge and Data Analysis step*. This step considers as input domain experts, data sources on consumer, nutrition, health and sustainability. It is possible to identify relevant data domains and conceptual entities forming the basis for modelling. The output of this step is a set of conceptual entities and relationships that guide ontology design and subsequent data preparation. ii) *Ontology Engineering step*. The process begins with the conceptual entities and domain relations defined in the previous step, which are collaboratively refined and expanded during the implementation of the FoCOSA ontology. It will provide the semantic backbone for the entire pipeline.

2. Engineering phase

The Engineering phase translates conceptual artefacts into computational components that integrate data, ontology, and LLM reasoning. This phase is articulated into two main steps: i) *Pipeline Design and Implementation step*. This step takes as input the FoCOSA ontology produced in the conceptual phase. Building on these artefacts, the process involves the design of prompt templates, the definition of retrieval strategies, and the creation of metadata-enriched chunks that preserve semantic context during indexing and retrieval. At this stage, the interactions between the LLM and the semantic resources are also specified, ensuring that the model can operate within ontological boundaries while leveraging contextual knowledge. The outcome of this step is an initial implementation blueprint of the pipeline, which connects the ontology modules to the retrieval and generation components, establishing the technical structure through which semantic grounding will be realised in practice. ii) *Integration and Reasoning step*. This step builds on the ontology-aligned data chunks and the pipeline design artefacts produced in the previous step. The process focuses on integrating vector-based retrieval with ontology-based reasoning, ensuring that semantic precision and contextual adaptability are jointly maintained. Retrieved chunks are aligned with FoCOSA entities, so that each piece of information used during generation remains consistent with the domain ontology. At the same time, LLM prompts are configured to embed ontology-derived constraints, guiding the model to generate outputs that respect the formal semantics of the domain. The result of this integration is a hybrid retrieval and reasoning pipeline in which each generated answer is semantically grounded, interpretable, and traceable to the underlying conceptual framework.

3. Evaluation phase

The Evaluation Phase assesses the consistency of the implemented system. As input, it will consider generated responses. To measure the retrieval performance, different metrics can be applied. This step focuses on the validation of the pipeline. The process involves iterative testing of retrieval accuracy and semantic alignment, followed by parameter adjustments to optimise coherence between the ontology-based representations and the LLM-driven retrieval and generation components. This step thus produces empirical and interpretive evidence that the system as a whole satisfies its design principles, establishing a link between conceptual formulation and observable results.

2.3. System architecture overview

The proposed system follows a modular, ontology-enhanced RAG architecture that realises the engineering phase of the methodological framework. The architecture is organised around the three standard RAG phases—Indexing, Retrieval, and Generation—with the FoCOSA ontology playing a guiding role at each stage. This design enables a clear separation of concerns between knowledge organisation, information access, and response synthesis, while maintaining semantic consistency through ontological grounding.

Indexing Phase: Knowledge Organisation. This phase transforms raw data into retrieval-ready representations. Data is segmented into chunks, and each chunk receives a semantic description generated using the FoCOSA ontology as a conceptual guide. This ensures that textual representations capture domain-relevant concepts before embedding. Chunks are then encoded as dense vectors and stored in separate collections (Food, Nutrition, Sustainability) that mirror the modular structure of the ontology. Importantly, the ontology itself is not embedded or stored in the vector database; it serves as a semantic reference for organising and describing content.

Retrieval Phase: Semantic Query Processing and Information Selection. This phase bridges natural language input with structured domain knowledge. First, the user's query is reformulated using ontological concepts and profile information to improve semantic alignment with indexed content. The ontology guides this query preprocessing step without requiring formal query translation (e.g., SPARQL). The reformulated query is then used to retrieve the most relevant chunks from the appropriate collection(s) in the vector database. The system adaptively targets single or multiple collections depending on whether the query spans one domain or multiple domains (e.g., combining nutrition and sustainability).

Generation Phase: Response Synthesis. This phase synthesises the final response by integrating retrieved evidence with ontology-derived semantic constraints. The LLM receives the user's query, the retrieved chunks, and structured ontology fragments as textual input. It produces conversationally fluent advice that respects the formal semantics defined by the ontology, ensuring consistency with nutritional requirements, health conditions, and sustainability constraints. This phase is responsible for personalization, multi-domain integration (e.g., balancing nutrition with environmental impact), and generating actionable recommendations.

Architectural positioning

From a GraphRAG perspective (discussed in Section 3.2), the architecture aligns with the *graph as a subject matter expert* pattern. FoCOSA, as the main source of contextual knowledge, is employed across all RAG phases rather than as a queryable database. Formal query languages (e.g., SPARQL, Cypher) are deliberately not used at runtime; instead, ontology fragments are supplied as structured textual input to the LLM, allowing it to reason over domain concepts using its native language-processing capabilities.

The concrete realization of this architecture, including component specifications, implementation details, and operational data flows, is presented in Section 6 (illustrated in Fig. 4).

3. Background on knowledge-integrated dialogue systems

This section provides an overview of the main foundational approaches relevant to our work, focusing on two research streams: (a) ontology-based methods for task-oriented dialogue systems (Section 3.1); (b) the integration of LLMs with Knowledge Graphs (KGs) in Retrieval-Augmented Generation (RAG) frameworks (Section 3.2). The development of modern dialogue systems increasingly leverages the aforementioned strategies to seamlessly combine structured and unstructured knowledge to enhance reasoning, reliability, and adaptability across diverse application contexts.

3.1. Ontology-based task-oriented dialogue systems

Ontology-based task-oriented dialogue systems utilize structured knowledge representations, such as ontologies and KGs, to provide semantic context and enable domain-specific reasoning. As a result, they enhance the system's ability to understand user intents, maintain coherent dialogues, and effectively support task completion within well-defined domains. Current work in this field has been focused on exploiting structured knowledge to advance dialogue state tracking, response generation, and domain adaptation. Ontology-aware pretrained language models have shown notable gains in end-to-end performance [17], as they embed domain ontologies directly in pretraining to better capture task-specific concepts and relationships.

In parallel, core ontologies designed for ontology-based dialogue systems in industrial applications have been developed, such as those within the TODO framework for Industry 4.0, which provides a modular architecture enabling natural interaction with complex industrial systems [18]. Additionally, the challenge of dynamically incorporating domain knowledge has been addressed through KG plug-in modules, which extract conversation-relevant information to guide response generation [19].

Recent approaches use ontology-based architectures to capture company expertise, combining NLP, domain ontologies, and KGs to expand knowledge bases through dialogue. For instance, Iga and Silaghi [20] propose a comprehensive architecture that facilitates domain-specific knowledge acquisition while managing multiple discussion threads, and demonstrate CRUD operations on KGs through natural conversation. While ontology-based task-oriented dialogue systems provide structured and interpretable reasoning, their rigidity and shallow inference capabilities have motivated recent research to explore how LLMs and KGs, within RAG frameworks, can enable more flexible and powerful reasoning over complex or unstructured knowledge, as discussed in the following section.

3.2. LLMs and knowledge graphs integration in RAG frameworks

The integration of LLMs within RAG frameworks has become a central strategy to combine generative language capabilities with domain-grounded knowledge. Although LLMs excel at understanding and producing natural language, they often lack up-to-date or specialised facts. RAG mitigates this issue by retrieving external sources (e.g., documents, passages, structured data) to guide the generation of responses. Typically, RAG pipelines operate in three phases (namely, indexing, retrieval, and generation), where source texts are encoded as embeddings in a vector database to augment the context of the LLM. However, relying solely on unstructured text can fragment document structure, return irrelevant results and fail to preserve the semantic relations and entity structures inherent in domain knowledge. This limitation has motivated the explicit inclusion of KGs and ontologies in the RAG pipeline, thus providing structured context and improving retrieval relevance, as shown by preliminary research efforts (e.g., OG-RAG [21]).

Building on these insights, the integration of KGs with LLMs into RAG pipelines paved the way to the *GraphRAG* paradigm, which has been investigated in various application domains [22]. Unlike traditional RAG systems based solely on plain text, GraphRAG uses a graph database as a source of contextual knowledge for the LLM, combining structured entities with textual descriptions, properties, and relations, thus enhancing Q/A over complex data and helping in hallucinations reduction. Surveys and white papers analyse GraphRAG approaches differently, distinguishing between those investigating how a KG is *operationally* used across the stages of the RAG pipeline [23] and other approaches studying what are the different *roles* a KG may assume within the overall RAG architecture. Below we adopt the latter viewpoint, also promoted by industry initiatives such as GraphWise,[1] which distinguishes three KG roles. The positioning of our approach with respect to such roles is discussed in Section 4.4.

(a) Graph as a content hub. In this variant, the graph functions as a content and metadata hub, integrating both domain entities and text chunks. Two main variants can be distinguished. In the first one, the graph directly incorporates chunks as nodes (storing their text, embeddings, and metadata) and connects them to the corresponding entity nodes. In the second variant, the graph maintains only metadata and identifiers that reference chunks and embeddings stored externally in a vector database. The RAPTOR framework exemplifies the graph as a content hub via recursive abstractive processing for tree-organised retrieval, where graph structures serve as hierarchical metadata [24]. Similarly, Microsoft GraphRAG uses graph structures to organise textual content through community hierarchies [25]. Neo4j, instead, has developed a Python library[2] supporting the GraphRAG paradigm through three strategies (naive, advanced, and modular RAG) at increasing degree of complexity for integrating LLMs with graph-based knowledge, offering reusable components, best practices, and ready-to-deploy workflows.

[1] https://graphwise.ai/fundamentals/what-is-graph-rag/.
[2] https://neo4j.com/docs/neo4j-graphrag-python/current/.

(b) Graph as a subject matter expert. In this variant, the graph acts as a "conceptual expert", providing structured semantic context to the LLM. For each query issued by the user, the system retrieves and supplies descriptions of entities, relationships and concepts (drawn, for instance, from ontologies, taxonomies, or domain-specific conceptual models) that enrich the LLM context for boosting the inference process. The graph focuses on structured and conceptual knowledge, while text chunks typically remain in an external vector database and are linked to the graph only for provenance or evidence when necessary. This approach enables explainable and multi-hop reasoning, but requires a well-curated KG. Think-on-Graph represents a sophisticated implementation of this approach, establishing deep reasoning methodologies that leverage KGs as domain experts to provide semantic context and conceptual relationships to LLMs [26]. In [27], the KG serves as a conceptual expert guiding the selection of relevant knowledge paths for multi-document question answering. The KG prompting approach presented in [28] further exemplifies the multi-document question answering, using graphs as semantic experts to enhance LLM reasoning capabilities. The EA-RAG approach for entity alignment [29] leverages the KG as a semantic expert, providing the generative model with structured context about entities and their relationships to support the alignment process.

(c) Graph as a database. In this approach, the KG is treated as a queryable collection of factual and metadata triples, serving as the primary source of information. Natural language questions are translated, either partially or fully, into executable graph queries (e.g., based on SPARQL or Cypher), which are then executed against the graph. The retrieved results are subsequently summarised or contextualised by the LLM. This approach focuses on factual triples and attributes that can be queried, with textual chunks acting as supporting evidence that can be stored separately or linked to relevant graph entities. This approach demands advanced natural language-to-graph query translation techniques and reliable entity disambiguation, but offers precise, grounded answers directly derived from the KG factual content. A plethora of practical implementations of this approach has been proposed. For instance, Cheng et al. [30] query graph structures directly and process the retrieved information with LLMs to support multi-step reasoning tasks. In the same vein, the Graphusion framework for scientific KG construction enables direct querying of factual information stored in graph databases [31]. Similarly, Xu et al. [32] map natural language queries to graph database operations in the scope of customer service question answering. Lastly, the study presented in [33] focuses on the impact of KG representations on question answering, where natural language questions are mapped into graph queries and the retrieved results are mediated by an LLM.

4. Related work

In this section, we examine domain-specific contributions in the food domain, focusing on three main directions: (a) the role of ontologies and KGs in structuring food-related knowledge (Section 4.1); (b) the use of LLMs in conversational food and dietary advisory systems (Section 4.2); (c) the application of RAG in food advisory systems to enhance information retrieval and contextual relevance (Section 4.3). Recent advancements in these areas highlight how ontologies, KGs, and LLMs jointly contribute to more accurate and context-aware food advisory systems. Finally, we close the section by discussing the positioning of our proposal with respect to the surveyed literature (Section 4.4).

4.1. Food-related ontologies and knowledge graphs

The development of ontologies in food-related domains has significantly advanced interoperability and data integration, with frameworks such as the OBO Foundry[3] supporting standardised vocabularies [34]. Among these, FoodOn [35] serves as a comprehensive "farm-to-fork" ontology, integrating terms from diverse sources, including Environment Ontology (ENVO) for environmental contexts, Agronomy Ontology (AgrO) for agronomic attributes, CDNO for dietary nutrition, and DO and MONDO for disease-related terminologies. Expanding on these efforts, FoodKG combines FoodOn with other ontologies, such as ChEBI, to link food, chemical properties, and nutritional data into a cohesive knowledge graph [2]. Despite these advancements in formal representation of food-related knowledge, significant gaps remain in addressing consumer-centric needs such as personalised dietary preferences, sustainability goals, individual health conditions, and contextual nutritional requirements. In fact, food advisory systems predominantly rely on generic algorithmic approaches that often either oversimplify dietary recommendations or create restrictive user profiles, thus limiting nutritional diversity and sustainable food choices, with these systems struggling to overcome such limitations [1]. The emerging integration of LLMs with ontological knowledge (e.g., [13]) presents a promising approach for overcoming these issues.

4.2. LLM-based food and dietary advisory systems

LLM-based approaches for food-related advice and recommendations have gained attention, as they use natural language understanding and generative abilities of LLMs to enhance interactions and capture implicit, context-dependent user preferences more effectively than traditional recommender systems. An interesting work in this area is FoodGPT [36], a LLM specifically designed for the food testing domain. It leverages incremental pre-training to incorporate domain-specific knowledge, e.g., from scanned documents, and uses a knowledge graph to mitigate hallucinations.

Some works propose fine-tuned LLMs for dietary analysis and suggestions. FoodSky [37] provides food recommendations by leveraging extensive Chinese food data and models trained to understand food semantics and generate context-aware suggestions

[3] https://obofoundry.org/.

for diverse dietary needs. Its features include personalised dietary advice and recipe generation. Similarly, the ROBERTA food recommendation system adopts a fine-tuned model on nutritional data and adapted to the food domain [38]. This system captures nuanced textual information related to dietary preferences, health profiles, and nutritional content to provide personalised and context-aware dietary suggestions, also aimed at promoting healthier eating choices. A novel deep generative network architecture is proposed in [39] to create weekly meal plans, enabling personalised nutrition. Leveraging LLMs, the AI-based recommendation system provides diverse, accurate suggestions for different user groups, with ChatGPT used to create a pool of alternative meals. Another notable framework, Personalization-Aware LLMs for Recommendation (PALR) [40], fine-tunes LLMs to rank food-related items based on user history and interactions, instructing LLMs to operate within a predefined candidate set.

Conversational recommendation systems incorporating LLMs have also been explored, with some of them leveraging graph-based reasoning techniques to model user preferences over time [41]. These systems enhance personalisation by dynamically adapting recommendations to evolving user interactions.

Other approaches investigate how to introduce multimodal interactions in LLM-based food recommendation. Chef Dalle [42] is a recipe recommendation system using multimodal interactions (voice, text, and images) via Whisper and GPT-Vision. It leverages conversational AI models to filter recipes based on user-provided ingredients and to deliver personalised suggestions. A system for suggesting meal preparations is Smart Cuisine [43], that reduces food waste by generating recipes using available ingredients and minimising leftovers. Similarly to our proposal, which considers multiple perspectives, MOPIHFRS [44] is a multi-objective food recommendation system that optimises user preferences, personalised healthiness, and nutritional diversity.

Despite their potential, LLM-based food and dietary advisory and recommendation systems face two main challenges. Firstly, expert supervision is required to ensure appropriate suggestions for users with specific health conditions, since LLMs may lack the depth needed for complex cases [45,46]. Secondly, LLMs struggle to integrate structured domain knowledge and are prone to hallucinations. Early work used ontology-based task-oriented dialogue systems (Section 3.1) to improve understanding and response generation, but these systems were costly to maintain and less adaptable. Research then moved towards approaches combining LLMs with KGs for Retrieval-Augmented Generation (Section 3.2). These latter systems maintain the reasoning advantages of ontology-based task-oriented dialogue while improving scalability, adaptability to new domains, and factual reliability.

4.3. Retrieval-augmented generation in food advisory systems

In food advisory systems, RAG facilitates evidence-based nutritional recommendations by providing information such as dietary guidelines and user preferences as contextual knowledge for generative models. Several recent studies illustrate the potential and flexibility of this approach.

The Pic2Plate system exemplifies the integration of Vision-Language Models (VLMs) with RAG for personalised recipe recommendations [47]. Pic2Plate employs an online retrieval mechanism that continuously updates its recipe database, enabling dynamic adaptation to newly available culinary information. Similarly, Liu et al. [48] proposed a Stochastic Diversified Retrieval Augmentation method to retrieve recipes semantically aligned with input images from existing databases. A Self-Consistency Ensemble Voting mechanism is used to select the most reliable recipe predictions by evaluating the consistency among generated candidates.

Beyond recipe retrieval, RAG has been leveraged also for clinical and nutritional applications. For instance, Kalpakoglou et al. [49] designed an AI-based nutrition recommendation system that retrieves dishes from an expert-validated Mediterranean food database through a structured four-step process, achieving 98% accuracy in energy intake and over 90% in macronutrient recommendations for generated user profiles.

4.4. Positioning and novel contribution

As we discussed, significant progress has been made in various aspects of systems providing food-related advice. However, there remains a gap in comprehensively integrating nutritional, sustainability, and personal preference factors within a single system. Moreover, the potential of combining advanced LLMs, offering a conversational interaction modality with domain-specific ontologies on food, is largely unexplored. Additionally, LLMs may require fine-tuning and integration with domain-specific knowledge bases to achieve optimal results. Our research addresses these gaps by integrating a modular ontology within a RAG-based system to enhance the contextual knowledge of LLMs, improving the accuracy and relevance of food advice while promoting sustainability and healthy eating habits.

Our approach lies at the intersection of traditional ontology-based task-oriented dialogue systems and GraphRAG paradigm. In fact, with respect to traditional ontology-based task-oriented dialogue systems, which are typically designed for narrow domains and rigid dialogue flows, we employ the ontology not as a backbone for dialogue state tracking, but as a semantic resource integrated across multiple phases of the RAG pipeline, thereby enhancing flexibility and conversational richness leveraging the generative capabilities of LLMs.

Recalling the GraphWise classification introduced in Section 3.2, our approach represents a configuration that primarily corresponds to the *graph as subject matter expert* role. Nonetheless, it differs from the standard formulation of this pattern, since it adopts a modular ontology, conceived as a formal conceptual model of the domain rather than an instantiated KG, as the main source of contextual knowledge consistently exploited across all RAG phases. Therefore, to emphasise this distinction, we refer to our approach as an *ontology-enhanced* GraphRAG pattern.

More specifically, the affinity with the *graph as subject matter expert* configuration stems from the role of the ontology as a curated conceptual expert that provides domain-specific knowledge to the LLM without embedding or storing textual content. Textual data are processed and indexed separately within a dedicated vector database, while the ontology supports semantic description generation for textual chunks during indexing, ontology-aware query rewriting during retrieval, and conceptually aligned response generation. Collectively, these design choices yield a pipeline in which generation-driven components actively enhance retrieval through generative reasoning, a core characteristic of Generation-Augmented Retrieval (GAR) systems [50].

At the same time, the approach clearly diverges from the *graph as content hub* pattern, since neither textual chunks nor their embeddings are stored in the ontology. In fact, the ontology serves only to guide the organization of the vector store into multiple embedding collections, each corresponding to a high-level conceptual module (e.g., food, nutrition, sustainability).

Finally, the proposal also distinguishes from the *graph as database* pattern, since the ontology is neither hosted in a triplestore nor accessed through formal query languages such as SPARQL or Cypher. In our framework, the ontology is maintained as a set of modular fragments and stored in a file (serialised in Turtle representation format). At runtime, when required during the RAG stages, it is provided directly to the LLM as structured textual input, enabling the model to reason directly over the conceptual structure using its native language-processing capabilities.

The integration of an ontology in the context of LLMs was also discussed in our earlier exploratory work [51], where contextual knowledge, such as domain files and ontology fragments, was injected directly into ChatGPT prompts. That preliminary approach relied solely on prompt engineering, without the use of RAG. Conversely, the current work introduces a robust ontology-based RAG framework in which, as previously noted, the ontology supports a bidirectional RAG process.

5. The FoCOSA ontology

The methodology adopted for engineering our ontology was mainly based on the Linked Open Terms (LOT) methodology [52], an industrial method for developing ontologies and vocabularies. The LOT methodology emphasises the reuse of terms (ontology classes, properties, and attributes) from already published vocabularies or ontologies. In the following, we introduce key aspects of how we used and adopted the methodological steps.

5.1. Requirements and competency questions

To develop the FoCOSA ontology,[4] we clarified the requirements by proposing Use Case (UC), Competency Questions (CQ) and additional restrictions (AR). During the process two knowledge engineers and one domain expert were involved.

Use case specification. We investigated specific use cases with the respective goals to be achieved by the ontology data modelling.

- *UC1: Health and nutritional monitoring.* This use case supports consumers who want to monitor and retrieve health parameters, dietary preferences, and physical activity levels to receive personalised nutritional and health advice. Both consumers and dietitians are involved in the monitoring and management of health and well-being. The use case flow starts from consumers who aim to receive food recommendations according to their health conditions and lifestyle preferences. Dietitians will take care to provide data on micronutrition and macronutrition and their correlations with physical conditions.
- *UC2: Sustainable shopping and consumption.* This use case focuses on analysing consumer shopping behaviour and providing recommendations aligned with dietary preferences and sustainability goals. Consumers, as well as different types of actors over the Food Supply Chain, are involved in the provisioning and exploitation of data on the sustainability of food products. According to the flow of this use case, consumers plan and perform purchases that can be tracked through shopping lists and carts. The ontology supports evaluating how past and planned purchases align with nutritional needs and sustainability targets, providing feedback for responsible consumption.
- *UC3: Product sustainability assessment.* This use case supports the comparison of food products and identification of sustainable alternatives based on environmental indicators. Consumers, product providers, and sustainability experts are involved in this use case. According to the flow of this use case, the system retrieves sustainability metrics (e.g., carbon and water footprint) associated with food items and compares alternative products. The ontology enables reasoning over sustainability indicators and suggests environmentally preferable substitutes.

Each use case contributes to define the ontology scope and ensures that the conceptual model captures essential relationships among consumers, food products, health parameters, and sustainability indicators.

According to the LOT methodology, the *Competency Questions (CQs)* defined under the ontology requirements phase represent the key queries that the ontology should be able to answer once developed. These questions are treated as part of the ontology requirement specification and guide its conceptualisation and formalisation. In this work, we adopt the LOT approach, where CQs are initially formulated in natural language as *informal competency questions*, since they are not yet expressed in a formal query language (e.g., SPARQL or Description Logic). This choice ensures that the ontology development process remains requirements-driven, while maintaining clear traceability between use cases, competency questions, and the corresponding ontology modules. For completeness and consistency with ontology engineering best practices, we also acknowledge complementary guidelines on CQ

[4] https://dbwis.gitlab.io/focosa.

Table 1
Examples of competency questions.

CQ	UC	Competency question
CQ1	UC1	How do specific food items impact various health conditions?
CQ2	UC1	What are the macronutrients and micronutrients present in a specific food item?
CQ3	UC1	What are the dietary restrictions of a specific consumer?
CQ4	UC1	What are the physical activities of a consumer?
CQ5	UC2	What food items are recommended for the consumer's weekly shopping based on their dietary preferences and sustainability goals?
CQ6	UC3	How do different food items compare in terms of sustainability?
CQ7	UC2	How do the consumer's past purchases align with water consumption?
CQ8	UC3	What are the sustainable alternatives for a specific food item?
CQ9	UC2	How do the consumer's past purchases align with her sustainability goals and how can she minimise waste?

formulation such as those proposed by Keet [53], which emphasise linguistic precision and formal mapping, even though our focus here is on engineering traceability through LOT.

We cooperated with the domain expert in the definition of both CQs. At the end of this stage, we provided the full list of competency questions to the domain expert, who performed its validation in terms of accuracy and completeness with respect to the identified use cases. The competency questions given in Table 1, with their relation to use cases, serve as the basis for validating the coverage and reasoning capabilities of the FoCOSA ontology.

Further, we propose a list of additional restrictions that helped in defining concepts and relations at the logical level.

- AR1: A food item can be associated with one or more health conditions that it may positively or negatively affect.
- AR2: Each food item can be described by a set of nutritional components, including macronutrients and micronutrients.
- AR3: A consumer can have dietary restrictions or health profiles that constrain the selection of suitable food items.
- AR4: A consumer can perform activities that influence nutritional requirements and food recommendations.
- AR5: A consumer can receive food recommendations that take into account both dietary preferences and sustainability goals.
- AR6: A food item can be compared to others according to sustainability indicators such as carbon or water footprint.
- AR7: Purchase information can be analysed in relation to environmental indicators, including water consumption and waste generation.
- AR8: Past purchase behaviour can be evaluated against declared sustainability goals to identify actions for improvement.

These additional restrictions ensure coherent reasoning across modules, while maintaining local independence within each domain.

5.2. Description of the FoCOSA ontology

Built using semantic technologies and aligned with the principle of reuse, FoCOSA integrates several concepts from other ontologies, such as the OBO Foundry food ontology [34], ensuring interoperability among them. The reuse of existing classes and properties is done using the `rdfs:subClassOf` or `owl:equivalentClass` construct for classes. The ontology consists of three modules: *Consumer*, *Nutritional* and *Sustainability*. They define around 41 core concepts, 30 object properties to capture semantic relations, and 18 datatype properties encoding measurable attributes such as nutrient, quantities, etc. The competency questions can be addressed through the concepts and relations defined in the three FoCOSA modules: CQ1–CQ4 are covered by the *Consumer* and *Nutritional* modules, CQ5, CQ7, and CQ9 by the *Consumer* and *Sustainability* modules, and CQ6 and CQ8 primarily by the *Sustainability* module. Each module of the ontology is described in detail in the following. Turtle fragments are provided in Appendix to illustrate the RDF/OWL implementation corresponding to the DL axioms presented for each module. To enhance readability and prevent repetition, all prefixes employed in the following section are summarised below in Listing 1.

```
@prefix focosa-c: <https://dbwis.gitlab.io/focosa/src/modules/consumer#> .
@prefix focosa-f: <https://dbwis.gitlab.io/focosa/src/modules/health-nutrition/food#> .
@prefix focosa-h: <https://dbwis.gitlab.io/focosa/src/modules/health-nutrition/health#> .
@prefix focosa-s: <https://dbwis.gitlab.io/focosa/src/modules/sustainability#> .
@prefix owl: <http://www.w3.org/2002/07/owl#> .
```

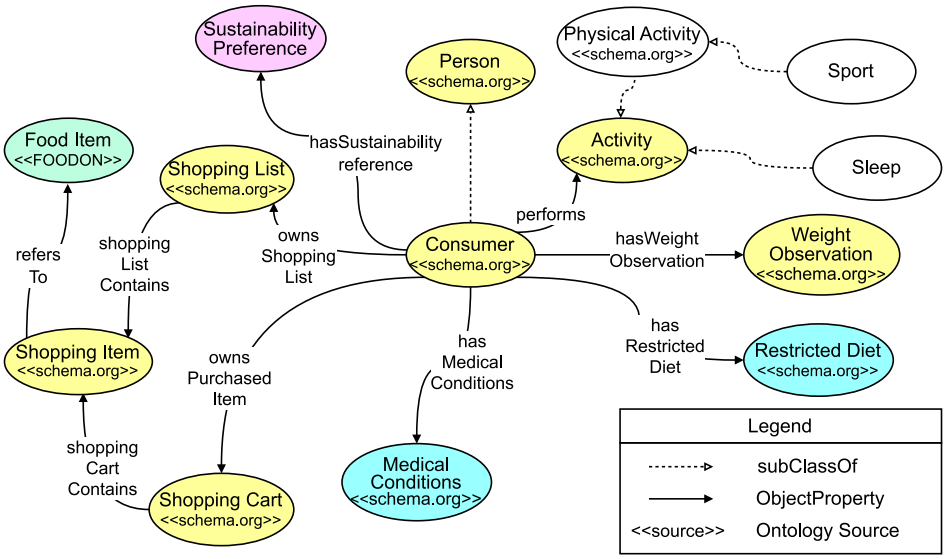


Fig. 1. Ontological concepts and relationships of the *Consumer* module in the FoCOSA ontology.

```
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
@prefix schema: <http://schema.org/> .
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .
@prefix foodon: <http://purl.obolibrary.org/obo/FOODON_> .
@prefix snomed: <http://purl.bioontology.org/ontology/SNOMEDCT/> .
```

Listing 1: FOCOSA ontology prefixes.

Consumer module

The *Consumer* module shown in Fig. 1, defines the top-level concepts and relations by capturing consumers' personal information, activities, observations, and shopping behaviour. It provides the basis for integrating nutritional and sustainability aspects. It reuses existing vocabularies such as **Schema.org**, **DBpedia**, and the **Time Ontology in OWL**. The complete OWL/Turtle implementation is available in the public repository.[5]

(a) Consumers and personal data. The class `Consumer` extends `schema:Person`, reusing personal attributes from Schema.org such as `address`, `birthDate`, and `gender`.

```
focosa-c:Person a owl:Class ;
    owl:equivalentClass schema:Person .

focosa-c:Consumer a owl:Class ;
    rdfs:subClassOf focosa-c:Person ;
    rdfs:comment "Represents someone who buys and consumes food
    items." .
```

Listing 2: Core classes and reuse of Schema.org.

(b) Activities. Lifestyle activities are represented through the class `Activity`, with subclasses `PhysicalActivity`, `Sleep`, and `Sport`. Time-related properties such as `hasDate`, `hasDuration`, and `hasFrequency` reuse the Time Ontology in OWL.

```
focosa-c:PhysicalActivity rdfs:subClassOf focosa-c:Activity ;
    owl:equivalentClass schema:ExerciseAction .

dbo:Sleep rdfs:subClassOf focosa-c:Activity ;
    owl:equivalentClass schema:Sleep .
```

[5] https://dbwis.gitlab.io/focosa/doc/consumer/.

```
dbo:Sport rdfs:subClassOf focosa-c:PhysicalActivity ;
    owl:equivalentClass schema:Sport .

focosa-c:hasDate a owl:DatatypeProperty ;
    rdfs:domain focosa-c:Activity ;
    rdfs:range xsd:date .

focosa-c:hasDuration a owl:DatatypeProperty ;
    rdfs:domain focosa-c:Activity ;
    rdfs:range xsd:integer .
```

Listing 3: Activities and alignments.

(c) Shopping habits. Shopping behaviour is modelled via `ShoppingList`, `ShoppingCart`, and `ShoppingItem`, aligned with `schema:ItemList`. Each cart or list contains items and can be linked to the date of purchase.

```
focosa-c:ShoppingList rdfs:subClassOf schema:ItemList .
focosa-c:ShoppingCart rdfs:subClassOf schema:ItemList .

focosa-c:shoppingCartContains a owl:ObjectProperty ;
    rdfs:domain focosa-c:ShoppingCart ;
    rdfs:range focosa-c:ShoppingItem .

focosa-c:shoppingListContains a owl:ObjectProperty ;
    rdfs:domain focosa-c:ShoppingList ;
    rdfs:range focosa-c:ShoppingItem .

focosa-c:shoppingDate a owl:DatatypeProperty ;
    rdfs:domain focosa-c:ShoppingCart ;
    rdfs:range xsd:date .
```

Listing 4: Shopping structures.

(d) Links to food items. Each `ShoppingItem` refers to a food item from the *Nutrition* module, represented by the property `refersTo`.

```
focosa-c:refersTo a owl:ObjectProperty ;
    rdfs:domain focosa-c:ShoppingItem ;
    rdfs:range focosa-f:FoodItem .
```

Listing 5: Link to food items.

Nutrition module

The nutrition module represents the intricate relationships between consumers, their dietary plans, and the various health and personal factors influencing their nutritional requirements. This module integrates key classes and their mutual relationships to encapsulate diverse aspects of nutrition and health. Fig. 2 shows this module with the main classes. It integrates established vocabularies such as **Schema.org**, **SNOMED CT**, and **FoodOn** to ensure semantic interoperability. The complete OWL/Turtle implementation is available in the public repository.[6]

(a) Medical conditions. The core class `MedicalCondition` captures general health-related aspects of the consumer. Specific subclasses such as `FoodAllergy` and `Pathology` provide a finer-grained representation of allergies and diseases, aligned with SNOMED CT.

```
focosa-h:MedicalCondition a owl:Class ;
    rdfs:label "Health Conditions" ;
    rdfs:comment "Represents health-related information." ;
    owl:equivalentClass schema:MedicalCondition .

focosa-h:FoodAllergy a owl:Class ;
```

[6] https://dbwis.gitlab.io/focosa/doc/health-nutrition/.

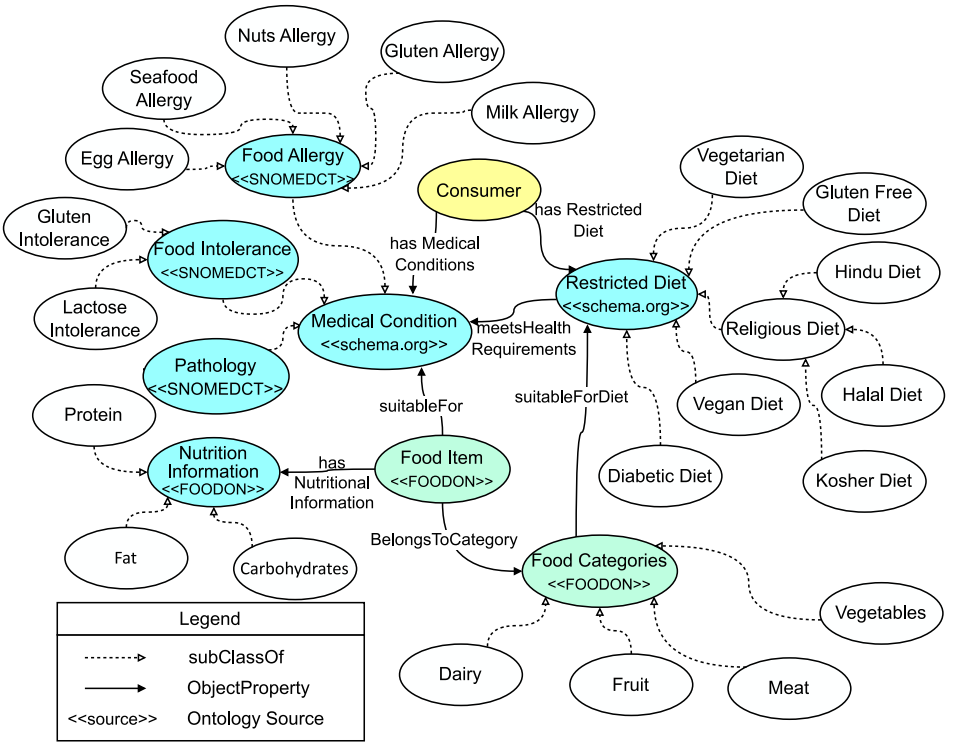


Fig. 2. Ontological concepts and relationships of the Nutrition module of the FoCOSA ontology.

```
    rdfs:label "Food Allergy" ;
    rdfs:subClassOf focosa-h:MedicalCondition ;
    owl:equivalentClass snomed:414285001 .

focosa-h:MilkAllergy a owl:Class ;
    rdfs:label "Milk Allergy" ;
    rdfs:subClassOf focosa-h:FoodAllergy ;
    owl:equivalentClass snomed:782555009 .

focosa-h:Pathology a owl:Class ;
    rdfs:label "Pathology" ;
    rdfs:subClassOf focosa-h:MedicalCondition ;
    owl:equivalentClass snomed:394595002 .
```

Listing 6: Medical conditions and alignments.

This hierarchy enables explicit specialisation of clinical conditions affecting diet recommendations (e.g., `Type2Diabetes` and `CardiovascularDisease` are modelled as subclasses of `Pathology`).

(b) Restricted diets. Dietary restrictions and preferences are represented through the class `RestrictedDiet`, specialised into specific categories such as vegetarian, gluten-free, or religious diets.

```
focosa-h:RestrictedDiet a owl:Class ;
    rdfs:label "Restricted Diet" ;
    owl:equivalentClass schema:RestrictedDiet ;
    rdfs:comment "Represents different types of restricted diets." .

schema:VegetarianDiet a owl:Class ;
    rdfs:label "Vegetarian Diet" ;
    rdfs:subClassOf focosa-h:RestrictedDiet .

schema:GlutenFreeDiet a owl:Class ;
    rdfs:label "Gluten Free Diet" ;
```

```
    rdfs:subClassOf focosa-h:RestrictedDiet .
```

Listing 7: Dietary restrictions and preferences.

These diet classes constrain which food items are compatible with a consumer's health and lifestyle needs.

(c) Food items and categories. The class `FoodItem` models individual food products and connects them to broader `FoodCategory` concepts defined in FoodOn. Each item can carry nutritional information and belong to one or more categories.

```
focosa-f:FoodItem a owl:Class ;
    rdfs:label "Food Item" ;
    rdfs:subClassOf foodon:00001002 .

focosa-f:FoodCategory a owl:Class ;
    rdfs:label "Food Categories" ;
    rdfs:subClassOf foodon:03400690 .

focosa-f:belongsToCategory a owl:ObjectProperty ;
    rdfs:domain focosa-f:FoodItem ;
    rdfs:range focosa-f:FoodCategory .

focosa-f:hasNutritionalInformation a owl:ObjectProperty ;
    rdfs:domain focosa-f:FoodItem ;
    rdfs:range focosa-f:NutritionInformation .
```

Listing 8: Food items and categories.

(d) Nutrients. Each food item may include information on macronutrients such as carbohydrates, fats, proteins, and omega-3. `NutritionInformation` has a quantitative value represented as a `xsd:float` through the `hasValue` datatype property.

```
focosa-f:NutritionInformation a owl:Class ;
    rdfs:label "Nutrition Information" .

focosa-f:Protein a owl:Class ;
    rdfs:subClassOf focosa-f:NutritionInformation .

focosa-f:hasValue a owl:DatatypeProperty ;
    rdfs:domain focosa-f:NutritionInformation ;
    rdfs:range xsd:float .
```

Listing 9: Nutrient information.

(e) Linking diets, health, and food. Relations connect dietary restrictions and medical conditions with suitable food items, enabling rule-based reasoning for compatibility checks.

```
focosa-h:meetHealthRequirements a owl:ObjectProperty ;
    rdfs:domain focosa-h:RestrictedDiet ;
    rdfs:range focosa-h:MedicalCondition .

focosa-h:suitableFor a owl:ObjectProperty ;
    rdfs:domain focosa-f:FoodItem ;
    rdfs:range focosa-h:MedicalCondition .

schema:suitableForDiet a owl:ObjectProperty ;
    rdfs:domain focosa-f:FoodCategory ;
    rdfs:range focosa-h:RestrictedDiet .
```

Listing 10: Linking diets, health conditions, and food items.

These relationships allow, for example, to infer that `BuckwheatPasta` (a `FoodItem`) is compatible with both `Gluten-FreeDiet` and `VegetarianDiet`, while a `SalmonFilet` would be excluded for a consumer affected by cardiovascular pathology.

Sustainability module

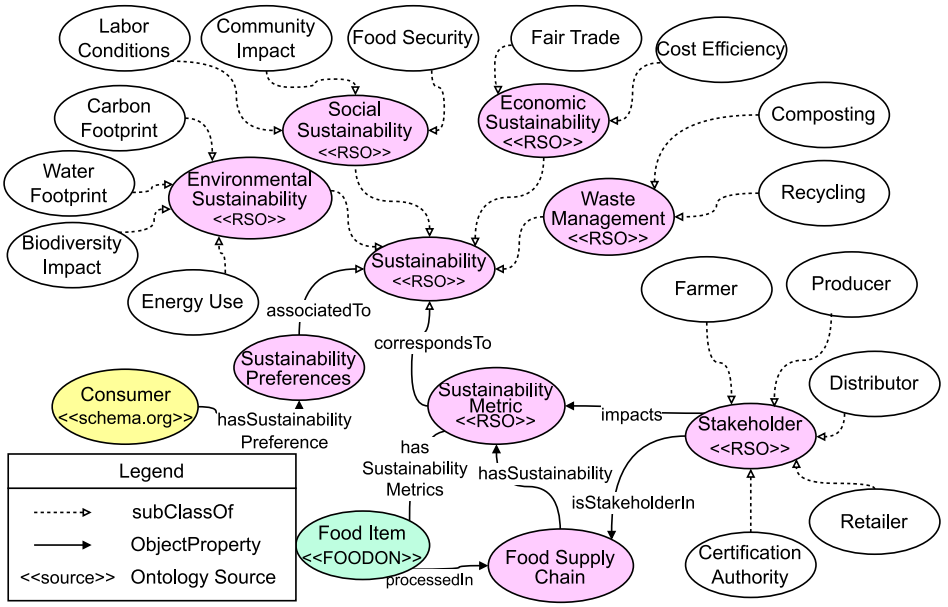


Fig. 3. Ontological concepts and relationships of the Sustainability module of the FoCOSA ontology.

The Sustainability module captures a comprehensive view of sustainability aspects within the food supply chain, emphasising the consumer's role in shaping sustainable practices. Fig. 3 shows that at the core of this module is the `Sustainability` class specialised into four subclasses. They serve to categorise the main areas of sustainable practices and impacts along the food supply chain. We rely on the RSO (Recipe Sustainability Ontology) vocabulary to model key concepts. It reuses concepts from the **Environmental Ontology (ENVO)** and the **AgroPortal** vocabulary. The complete OWL/Turtle implementation is available in the public repository.[7]

(a) Sustainability dimensions. The ontology distinguishes four complementary perspectives of sustainability, environmental, social, economic, and waste management, each represented as a subclass of `Sustainability`.

```
focosa-s:Sustainability a owl:Class ;
    rdfs:label "Sustainability" ;
    rdfs:comment "Represents the overall sustainability concept,
    including environmental, social, economic, and waste dimensions." .

focosa-s:EnvironmentalSustainability a owl:Class ;
    rdfs:label "Environmental Sustainability" ;
    rdfs:comment "Reflects ecological impact such as carbon footprint
    and resource use." ;
    rdfs:subClassOf focosa-s:Sustainability .

focosa-s:SocialSustainability a owl:Class ;
    rdfs:label "Social Sustainability" ;
    rdfs:comment "Captures social aspects such as fair trade
    and labor conditions." ;
    rdfs:subClassOf focosa-s:Sustainability .

focosa-s:EconomicSustainability a owl:Class ;
    rdfs:label "Economic Sustainability" ;
    rdfs:comment "Addresses economic aspects such as profitability
    and cost structures." ;
    rdfs:subClassOf focosa-s:Sustainability .

focosa-s:WasteManagement a owl:Class ;
```

[7] https://dbwis.gitlab.io/focosa/doc/sustainability/.

```
    rdfs:label "Waste Management" ;
    rdfs:comment "Covers reduction, recycling, and reuse of waste." ;
    rdfs:subClassOf focosa-s:Sustainability .
```

Listing 11: Core sustainability dimensions.

(b) Sustainability metrics. Measurable indicators such as carbon or water footprint, and compliance with labour practices, are modelled as subclasses of `SustainabilityMetric`.

```
focosa-s:SustainabilityMetric a owl:Class ;
    rdfs:label "Sustainability Metric" ;
    rdfs:comment "Represents measurable aspects of sustainability
    performance of food items, processes, or stakeholders." .

focosa-s:CarbonFootprint a owl:Class ;
    rdfs:label "Carbon Footprint" ;
    rdfs:subClassOf focosa-s:SustainabilityMetric .

focosa-s:WaterFootprint a owl:Class ;
    rdfs:label "Water Footprint" ;
    rdfs:subClassOf focosa-s:SustainabilityMetric .
```

Listing 12: Sustainability metrics.

(c) Links between consumers, food items, and sustainability metrics. Object properties connect consumer preferences, food items, metrics, and validating organisations, enabling reasoning on the alignment between sustainability goals and consumer behaviour.

```
focosa-s:hasSustainabilityPreference a owl:ObjectProperty ;
    rdfs:label "has sustainability preference" ;
    rdfs:domain focosa-c:Consumer ;
    rdfs:range focosa-s:SustainabilityPreferences .

focosa-s:hasSustainabilityMetric a owl:ObjectProperty ;
    rdfs:label "has sustainability metric" ;
    rdfs:domain focosa-f:FoodItem ;
    rdfs:range focosa-s:SustainabilityMetric .

focosa-s:impacts a owl:ObjectProperty ;
    rdfs:label "impacts" ;
    rdfs:domain focosa-s:Stakeholder ;
    rdfs:range focosa-s:SustainabilityMetric .
```

Listing 13: Relationships among consumers, food items, and sustainability metrics.

While the ontology itself defines the conceptual links between consumers, preferences, and certified sustainability metrics, these relationships can be exploited in reasoning or validation tasks to assess the alignment between consumer preferences and product sustainability profiles.

6. Ontology-enhanced RAG architecture

This section presents in detail the technical implementation of the ontology-enhanced RAG architecture,[8] whose logical design was outlined in Section 2.3. Fig. 4 illustrates the complete system architecture with its operational components, data flows, and interactions. The implementation follows the Advanced RAG pattern [54], which focuses on introducing operational steps apt to improve the retrieval quality. The system has been developed using LlamaIndex,[9] a renowned Python framework for coding LLM-based applications, with a vector database for embedding storage and retrieval. We now describe each component and phase in detail.

[8] The source code of the system is available at: https://gitlab.com/dbwis/food-rag-system/-/tree/DKE_2026.
[9] https://www.llamaindex.ai/.

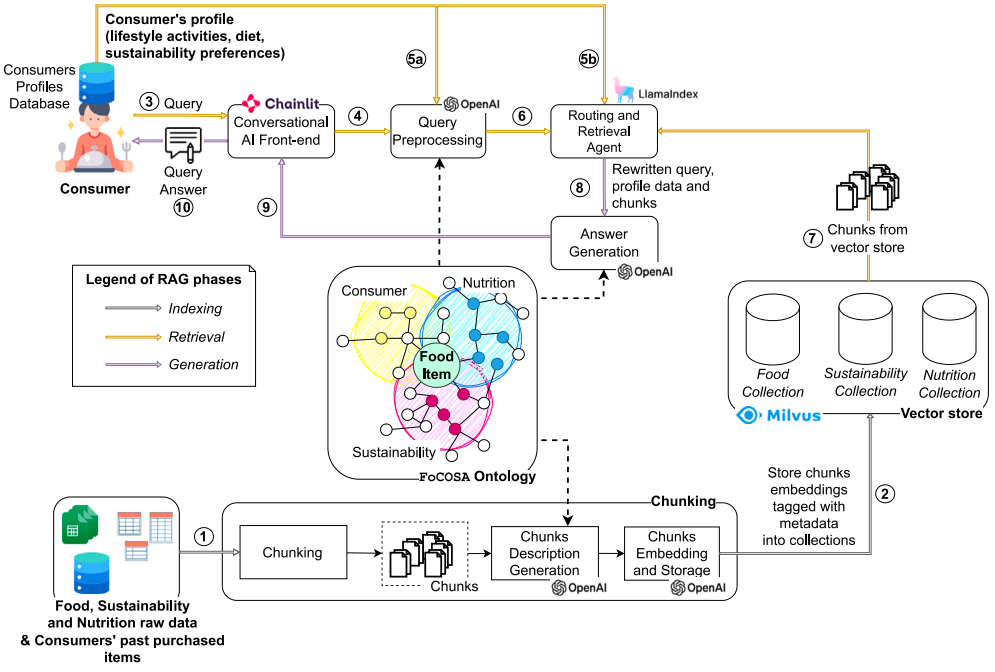


Fig. 4. Proposed ontology-enhanced RAG architecture of the conversational food advisory system. In the figure, three dashed arrows highlight the steps of RAG where the FoCOSA ontology has been employed.

6.1. *Indexing phase*

The first phase of the RAG approach focuses on transforming structured and unstructured raw data regarding food, nutrition and sustainability aspects into a format that supports efficient retrieval. To this aim, and to address also the potentially large size of datasets, they are segmented into *chunks* to maintain a balance between preserving contextual integrity and excluding irrelevant content (①). For each chunk, the *Chunks Description Generation* component leverages the `gpt-4o-mini` model from OpenAI and the FoCOSA ontology to produce a textual description of the chunk semantically aligned with the ontology (using the prompt template in Box 1). Then, each chunk is then encoded into a dense vector representation using an *embedding model* (specifically, the `text-embedding-3-small` model from OpenAI); the generated description is stored as metadata associated with the corresponding embedding and ingested into the Milvus[10] vector database by the *Chunks Embedding and Storage* component (②). Notably, according to this strategy, the FoCOSA ontology itself is not embedded into the vector database. Instead, it is stored in a dedicated file (serialised in Turtle format) and provided as a textual input to the LLM (through the `{ontology}` placeholder in the prompt template in Box 1) to guide the generation of ontology-aligned chunk descriptions. When embeddings are stored in the vector database, they are organised into *collections*, which serve as the primary organisational units to group embeddings and associated metadata. In our architecture, the *Chunks Embedding and Storage* component ensures that embeddings are stored in the appropriate collection, depending on the ontology module. Indeed, different collections have been employed to store the embeddings of chunks associated with food, nutrition, and sustainability data, thus reflecting the subdivision into modules of the FoCOSA ontology. This subdivision strategy ensures that the collections are optimally organised for downstream RAG phases. Noteworthy, only consumers' past purchases data are indexed in the vector store, while profile data (e.g., lifestyle activities, dietary requirements, and sustainability preferences) is not. This choice ensures that the retrieval process remains focused on broader, reusable knowledge rather than individual user-specific information.

[10] https://milvus.io.

Box 1 - Chunk Description Generation Template

You are a very proficient assistant capable of generating concise and semantically accurate descriptions (i.e., summaries) of input texts. Your task is to produce a description that aligns with the given ontology, ensuring that the output is both conceptually precise and contextually relevant. The description should focus on the following key aspects:

(1) Nutritional Information: (a) provide a breakdown of macronutrient content (carbohydrates, proteins, fats) and micronutrient content (vitamins, minerals) for the given food items; (b) highlight any notable nutritional characteristics, such as high concentrations of specific nutrients or suitability for particular dietary preferences (e.g., low-sugar, high-protein).

(2) Sustainability and Environmental Impact: (a) assess the sustainability metrics of each food item, including water usage, land use, and greenhouse gas emissions; (b) compare environmental impacts, identifying which items have lower ecological footprints or higher resource efficiency relative to others.

Ensure that the description is clear, structured, and directly aligned with the provided ontology while maintaining brevity and relevance.

Text: `{text}`

Ontology: `{ontology}`

Description:

6.2. Retrieval phase

In the retrieval phase, the system extracts the most relevant chunks from the pre-indexed collections based on the user's query, which is submitted through the Chainlit[11] *Conversational AI front-end* (③). The user's query is firstly rewritten through an LLM-based preprocessing step by the *Query Preprocessing* component (④), receiving as input the FoCOSA ontology as well as user's profile data regarding lifestyle activities, dietary requirements and sustainability preferences (5a). The aim is to produce a version of the query that is semantically aligned both with the domain knowledge from the FoCOSA ontology and with relevant information from the user's profile (using the prompt template in Box 2). In this respect, the refined version of the query better aligns with the strategy used to organise indexed data (which adheres to the modules of the FoCOSA ontology), thereby improving retrieval accuracy while ensuring the retrieval of chunks consistent with the aforementioned user's profile information. After refinement, the rewritten query is forwarded to the *Retrieval and Routing Agent* component (⑥), which implements an agentic routing method based on the *RouterRetriever* of LlamaIndex[12] for accessing and retrieving chunks from the vector store collections, so that the subsequent generation phase receives the most contextually relevant information. The routing agent combines the RouterRetriever with a LLM-based collection selector mechanism on the query reformulated by the *Query Preprocessing* component. Depending on the selector configuration, routing may target a single collection or multiple collections when the query spans more than one FoCOSA domain. Effects of the aforementioned configurations will be discussed in Section 7.2.1. After collection(s) selection, the rewritten query is encoded with the same embedding model used at indexing, cosine similarities are computed between the query vector and the vectors in the chosen collection(s), and retrieval returns the top-K (default $K = 5$) chunks from each collection (⑦). This value is a renowned trade-off fostered in literature to balance relevance with retrieval latency. Moreover, it helps preventing the so-called *long-context issue* [55], which may lead the LLM to focus on less relevant chunks in the generation phase of RAG or, in the worst cases, to the saturation of tokens that the LLM can process in a single pass. We remark that the RouterRetriever adheres to an agentic pattern since, while the deterministic aspect is enforced by top-K cosine similarity retrieval within the selected collection(s), the stochastic aspect arises from the LLM decision on which collections to consult, introducing flexibility and semantic adaptation to the user's query.

[11] https://docs.chainlit.io/get-started/overview.

[12] A retriever is an interface wrapping data sources with the aim of retrieving query-relevant data (in our case, the data sources are the collections of the vector store). Details on the *RouterRetriever* available at: https://developers.llamaindex.ai/python/examples/retrievers/router_retriever/.

Box 2 - Query Rewriting Template

You are an expert assistant that rewrites user queries into clearer, ontology-aligned natural language questions suitable for retrieval in a RAG system.

GOAL
(1) Preserve the original intent of the query.
(2) Align wording with the provided ontology concepts.
(3) Integrate consumer profile details (lifestyle activities, dietary restrictions, sustainability preferences) when relevant.
(4) Ensure the rewrite is textual, semantically faithful, and retrieval-friendly.
(5) Return only one final rewritten query as plain text (no JSON, no explanations).

INSTRUCTIONS
(1) If the query mentions a specific consumer (e.g., "Jason Brown"), include their dietary restrictions, sustainability preferences, and relevant lifestyle activities.
(2) If the query relates to shopping or purchase data, include relevant details (items, quantities, dates, nutritional information).
(3) If the query concerns food products, use ontology-aligned labels and clarify attributes such as nutrients, health impacts, or categories.
(4) If the query involves sustainability, specify the relevant metrics (e.g., carbon footprint, water usage, environmental impact).
(5) Keep the rewritten query close to the original question in style while making it precise and retrieval-ready.
(6) Output only the rewritten query as plain text, with no explanations or alternatives.

EXAMPLES
— 1 —
Original Query: "How do Andrew Chen's beverage purchases relate to his fair trade and water usage sustainability goals?"

Ontology (excerpt, illustrative):
- Class: FoodItem
- Property: hasSustainabilityMetric
- Individual Metrics (SustainabilityMetric): FairTradeScore, WaterUsageIntensity
- Property: purchasedBy (Consumer → FoodItem)

Rewritten Query: "How do Andrew Chen's purchases of beverages align with the sustainability metrics FairTradeScore and WaterUsageIntensity?"

— 2 —
Original Query: "What are the nutrient benefits of quinoa for people with diabetes?"

Ontology (excerpt, illustrative):
- Class: FoodItem (e.g., FoodItem:Quinoa)
- Class: Nutrient (e.g., Protein, Fibre, Magnesium)
- Class: HealthCondition (e.g., Diabetes)
- Property: containsNutrient (FoodItem → Nutrient)
- Property: hasHealthImpact (FoodItem → HealthCondition)

Rewritten Query: "What nutritional benefits (e.g., protein, fibre, magnesium) of quinoa are relevant for individuals with the diabetes?"

Ontology: `{ontology}`

Original query: `{original_query}`

User's profile data (lifestyle activities, dietary requirements, sustainability preferences): `{profile_data}`

Rewritten query:

6.3. Generation phase

The final phase of our RAG framework leverages an LLM (specifically, `gpt-4o-mini` model from OpenAI) to generate food advice for the user's query by exploiting the inference capabilities of the LLM upon the data retrieved from the previous RAG phase. The prompt submitted to the LLM (using the prompt template in Box 3), apart from containing a preamble devoted to

set the utterance and instruct the LLM for the generation of food advice, contains placeholders which are filled by the *Answer Generation* component with the following data, received in turn from the *Routing and Retrieval Agent* component (⑧): (i) the rewritten query (⑥); (ii) user's profile data (5b); (iii) the top-K chunks retrieved in the RAG Retrieval phase (from ⑦), which will be incorporated into the `{retrieved_data}` placeholder of the prompt. The *Answer Generation* component invokes the LLM with the aforementioned prompt, which instructs the LLM to produce an output answer semantically aligned with the FoCOSA ontology. This ensures that food advice adhere to the conceptualization and the terminology from the FoCOSA ontology (⑨). Lastly, the output query answer is forwarded to the Conversational AI front-end, to be inspected by the user (⑩).

Box 3 - Query Answering Template

You are a very proficient assistant in providing accurate, context-aware responses based on a user's query (for instance, a consumer willing to get food-related advice, based on his/her profile data). Your task is to answer the following query by considering the user's profile data, retrieved data from the vector store (also referred to as context), and the conceptual framework defined by the ontology.

INSTRUCTIONS:
(1) Ensure that your answer is relevant to the query and fully takes into account the user's profile data and retrieved data from the vector store.
(2) Align the response with the concepts and terminology of the provided ontology, to maintain semantic consistency.
(3) Provide a concise and accurate answer that is contextually appropriate and tailored to the user's needs.
(4) If the provided data is insufficient, please reply saying that the information is not enough to fully answer the query.

Query: `{rewritten_query}`

User's profile data (lifestyle activities, dietary requirements, sustainability preferences): `{profile_data}`

Retrieved data from the vector store: `{retrieved_data}`

Ontology: `{ontology}`

Answer:

7. Experimental evaluation

In this section, we perform an in-depth evaluation of our LLM-based approach. Specifically, we will focus on assessing the quality of the answers given by the conversational food advisory system, obtained downstream the Generation phase in the RAG architecture illustrated Section 6. To evaluate the quality of the LLM-generated answer, there are in literature two main viable strategies: (i) *human-based evaluation*, where humans are in charge of evaluating the LLM-generated text, providing scores to claims, based on their personal judgement, apt to capture quality aspects of the generated text; (ii) *LLM-as-a-Judge evaluation* [15], a strategy recently proposed to assess the output of an LLM (referred to as *evaluated LLM*), adopting an *evaluator LLM* (different from the evaluated LLM and offering superior inference performance), whose aim is to score the output of the evaluated LLM. The former strategy is commonly employed in complex domains, where the human judgement is crucial. The latter strategy, on the other hand, is typically used when humans are limited in number or unavailable to evaluate the LLM-generated output, thus providing an alternative which goes towards the setup of an automated quality evaluation process. In this paper, we employed the LLM-as-a-Judge approach to evaluate the quality of the answers generated by our conversational food advisory system. As part of future work, we plan to conduct also a human-based evaluation (with food domain experts) on the output generated by the RAG system proposed here, comparing the results with those obtained using the LLM-as-a-Judge strategy.

7.1. Experimental setup

7.1.1. LLM-generated responses evaluation criteria

The quality of LLM-generated responses has been evaluated using a comprehensive framework of criteria (Table 3). Noteworthy, we distinguish between: (i) *general criteria*, apt to assess if the LLM-generated response is accurate, understandable, logically consistent, and directly addresses user's query; (ii) *context-related criteria*, focusing on assessing the extent to which the LLM exploits the context provided as part of the prompt to generate the answer and whether hallucinations occurred or not. Criteria such as

Table 2
Datasets employed for the experimental evaluation.

Dataset name	Description	Provider	URL
food.csv	Comprehensive data about different types of food	CORGIS	Link
nutrition.csv	Nutritional values for common foods and products	Kaggle	Link
generic-food.csv	General information about various food items	GitHub	Link
food-production.csv	Environmental impact of food production	Kaggle	Link
customer_activities.csv	Data about consumer activities	N/A (Generated)	Link
detailed-dietary-requirements.csv	Detailed dietary requirements for various consumers	N/A (Generated)	Link
detailed-sustainability-preferences.csv	Detailed sustainability preferences of consumers	N/A (Generated)	Link
purchase_data.csv	Data about past purchases made by consumers	N/A (Generated)	Link
complete.owl	OWL representation of the complete FoCOSA ontology	GitHub	Link

Table 3
Criteria for evaluating the quality of the LLM-generated responses.

General criteria	
Relevance	Does the answer directly address the question?
Clarity	Is the response clear, coherent, and easy to understand?
Completeness	Does the answer cover all important aspects of the question?
Fluency	Is the language used in the response grammatically correct and natural?
Consistency	Is the response logically consistent with no contradictions?
Actionability	Does the response provide actionable advice (i.e., practical, useful, and feasible to follow in real life)?
Context-related criteria	
Minimisation of Hallucinations (MoH)	Does the LLM fabricate facts or suggest unsupported claims, which cannot be inferred from the provided context?
Data Utilisation Effectiveness (DUE)	How effectively a response leverages the available data in the context, to provide an informed and comprehensive response?

Relevance, *Clarity*, *Completeness*, *Fluency*, *Consistency* and *Minimisation of Hallucinations* (in brief, MoH) have been defined in previous research efforts as pivotal [56–58], and serve as a foundation for our quality evaluation assessment. Instead, the *Actionability* criterion assesses whether the LLM-generated response offers practical and feasible food advice, that the user can realistically follow. Although it is not a standard criterion in a LLM-as-a-Judge evaluation strategy, Actionability aligns with the notion of "Helpfulness" or "Instruction-Following" proposed in recent works [15,59]. As AI-based task-oriented systems, such as food advisory platforms, become more and more common, Actionability is increasingly regarded as a key criterion in ensuring that AI-generated advice is both practical and easy to follow by the user. To assess how effectively the retrieved information (which, in our case, may span multiple domains such as sustainability, nutrition, consumer habits) is exploited by the LLM to generate the response, in a way that contributes to deliver a complete, meaningful, and user-relevant answer, we introduced *Data Utilisation Effectiveness* (in brief, DUE) as a context-related evaluation criterion.

To operatively conduct the LLM-as-a-Judge evaluation, we adopted a GPT-4-based setup in which `gpt-4o-mini` served as the Evaluated LLM (temperature = 0.2, to balance diversity and focus, with a token limit of 512 to ensure concise yet informative responses), while `gpt-4o` acted as the Evaluator LLM (temperature = 0.0, to enforce deterministic and consistent judgements, minimising evaluation variability). Regarding the scoring methodology, each criterion was rated by the Evaluator LLM on a discrete scale, ranging from 0 to 5, where 0 indicates insufficient alignment with the criterion and 5 indicates optimal alignment with the criterion. To perform the assessment and obtain the score for each criterion, the prompt submitted to the Evaluator LLM included

Table 4
Queries for the quality assessment, classified according to their domains (i.e., covering modules of the FoCOSA ontology).

ID	Query description	Involved domain(s)
Q1	How does salmon impact the health of consumers with cardiovascular conditions?	Consumer, Nutrition
Q2	What macronutrients and micronutrients are present in salmon?	Nutrition
Q3	What are the dietary restrictions of Elizabeth Gonzalez, and what do you suggest she consume?	Consumer, Nutrition
Q4	What are the physical activities of Elizabeth Gonzalez, and what do you suggest she consume?	Consumer, Nutrition
Q5	What food items are recommended for Julie Chavez's weekly shopping based on their dietary preferences and sustainability goals?	Consumer, Shopping Habits, Nutrition, Sustainability
Q6	How do Pig Meat and Poultry Meat compare in terms of sustainability?	Sustainability
Q7	How do Julie Chavez's past purchases align with water consumption?	Shopping Habits, Sustainability
Q8	What are the sustainable alternatives for beef?	Sustainability
Q9	Do Sara Rogers' past purchases align with her sustainability goals and how can she minimise waste?	Shopping Habits, Sustainability

the textual definition of the criteria as from Table 3. Additionally, the prompt was supplemented with ad-hoc instructions to produce a textual justification, to ensure transparency in the evaluation and enable understanding of the rationale behind each decision.

As anticipated in Section 1, existing RAG evaluation frameworks, such as RAGAS [14], provide metrics that are not explicitly designed to evaluate the overall effectiveness of information use from the user's perspective in personalised, advisory scenarios. In particular, several RAGAS metrics (for example, *Context Recall* and *Noise Sensitivity*) require ground-truth information, which poses practical challenges. Indeed, while expert consensus could in principle establish a reliable ground-truth, doing so would require substantial expert involvement and resources that are not yet available at the current stage of our research. Similarly, programmatic approaches to generating gold labels (e.g., heuristic-based or automated pseudo-labelling strategies) would also be problematic, as they would likely introduce systematic bias. This limitation calls for alternative evaluation strategies, such as our LLM-as-a-Judge approach, which does not require ground-truth information and relies on the aforementioned general and context-specific criteria (Table 3), specifically designed to evaluate complex, open-ended generative outputs in domain-specific advisory systems, such as the one presented in this paper. Even though some RAGAS components (e.g., *AspectCritic*[13]) allow a certain degree of customisation, they produce binary judgements (i.e., alignment vs. non-alignment with a predefined aspect). Conversely, our evaluation strategy employs a multi-level ordinal scale (0–5) to produce graded, interpretable evaluations of LLM-generated advice. To further increase interpretability, we also require the LLM not only to assign a score but also to provide a brief justification for each rating. This approach highlights incremental improvements across system configurations and making judgements more interpretable than a simple binary assessment. A RAGAS-based evaluation is left to future work, once expert-validated ground-truth data become available.

7.1.2. Single-domain and multi-domain queries

To evaluate the quality of the LLM-generated responses, we prepared a set of nine queries (Table 4) that address diverse aspects of the food domain and that descend from the Competency Questions at the basis of the FoCOSA design methodology, described in Section 5. Such queries have been issued over the datasets in Table 2 and are classified, based on their complexity, in simple (*single-domain*) and complex (*multi-domain*) queries. Simple queries are domain-specific and require retrieval from (and inferencing on) a single dataset or ontology module (Q2, Q6, and Q8). Conversely, complex queries involve multiple ontology modules, and require integrating data spanning multiple datasets. Indeed, these queries often combine sustainability, health, and user-specific preferences/habits (Q1, Q3, Q4, Q5, Q7, Q9). Testing the conversational food advisory system with these two groups of queries ensures diverse evaluation scenarios, highlighting the ability of the system to handle both straightforward and articulated users' queries.

7.1.3. Ablation study configurations for testing the RAG architecture

Resembling an ablation study, different *configurations* have been devised for running tests over the RAG architecture of the conversational food advisory system (C1 to C5 in Table 5). Roughly speaking, a configuration determines whether the FoCOSA

[13] https://docs.ragas.io/en/stable/concepts/metrics/available_metrics/general_purpose/#aspect-critic.

Table 5
Configurations for the ablation study on the RAG architecture.

ID	Configuration name	Description	Retrieval mechanism (*selector*)
C1	No Ontology (Baseline)	The ontology is not used in any of the RAG phases.	Single selector, Multi selector
C2	Ontology-Enhanced Indexing	The ontology is employed only in the prompt used to generate chunks descriptions, before chunks embedding.	
C3	Ontology-Enhanced Retrieval	The ontology is employed only to perform query rewriting before retrieval from the vector store.	
C4	Ontology-Enhanced Response Generation	The ontology is employed only in response generation as additional contextual knowledge to formulate the answer to a given user's query.	
C5	Full Ontology	The ontology is employed in all RAG phases (indexing, retrieval, and response generation).	

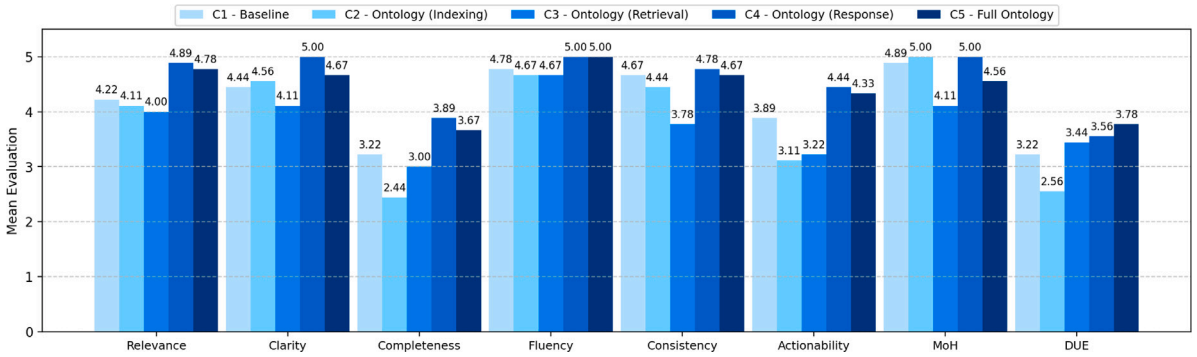


Fig. 5. Average quality criteria scores over the different configurations C1 to C5 (Single Selector).

ontology has been used or not in the different phases of RAG (*indexing*, *retrieval*, and response *generation*). For each configuration, two distinct retrieval mechanisms were evaluated for the *Routing and Retrieval Agent* component shown in Fig. 4. These retrieval mechanisms (referred to as *selectors*) are provided by the LlamaIndex framework[14] and are apt to ensure efficient query routing and data access tailored to query complexity. Indeed, one of the aspects we aim to assess is the impact a selector has on the outcome of a query, depending on its complexity (i.e., whether it is single-domain or multi-domain). In particular, two selector mechanisms have been employed in the *Routing and Retrieval Agent* component: (i) *PydanticSingleSelector* (in brief, Single Selector), identifies a single collection within the vector store to send the query, ensuring efficiency and precision for simpler domain-specific queries; (ii) *PydanticMultiSelector* (in brief, Multi Selector), meant for more complex queries requiring multi-domain integration, enables simultaneous retrieval from multiple collections from the vector store, being suitable for queries covering several domains, such as combining sustainability metrics with nutritional data. Overall, the number of chunks retrieved by each selector may be controlled through an hyperparameter (namely, `TOP_K_CHUNKS`, which is set by default to a value of 5, as explained in Section 6).

7.2. Results and discussion

In this section, we summarise the results of the experimental evaluation, focusing on different perspectives: (i) at a fine-grained level, analysing the effects of configurations on both general and context-related criteria for evaluating the quality of LLM-generated responses, when considering sample queries from Table 4; (ii) at a broader level, providing general insights into the impact that the different configurations have on the RAG architecture of the conversational food advisory system.

7.2.1. Impact of configuration variations on the evaluation of output quality

Considering general quality evaluation criteria, the ablation study conducted over the RAG architecture revealed that Clarity, Fluency and Consistency were the least affected by the adopted configuration (as illustrated in Figs. 5 and 6). This is due to the fact that, when dealing with LLMs, these criteria are somehow inherently concealed in the language model capabilities. Indeed, LLMs are designed to generate coherent and well-structured text, and their Clarity, Fluency and Consistency are typically enforced through

[14] https://docs.llamaindex.ai/en/stable/module_guides/querying/router/.

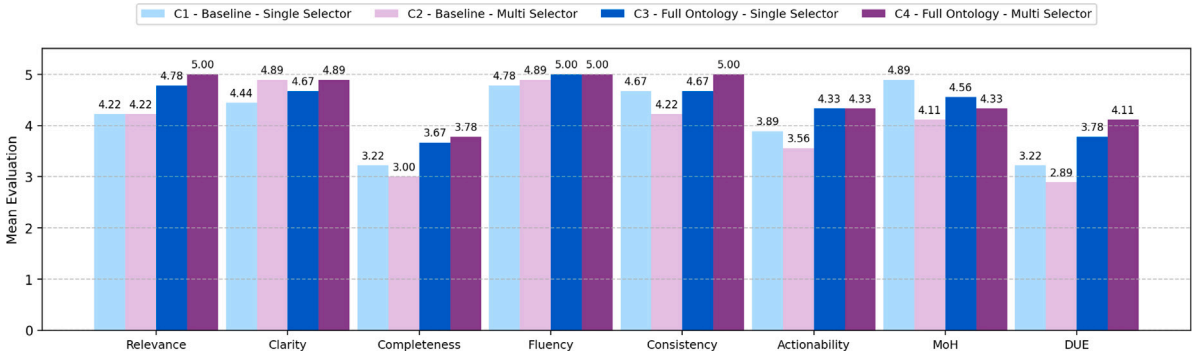


Fig. 6. Average quality criteria scores by selector type for C1 and C5 configurations.

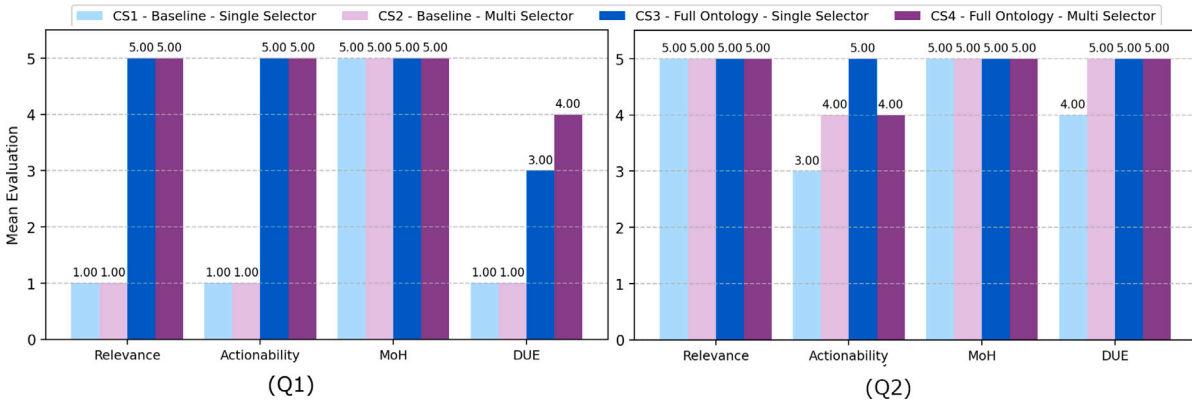


Fig. 7. Excerpt of quality criteria scores detailed for query Q1 (left) and Q2 (right).

specific prompt instructions (see the prompt template for query answering in Section 6). Conversely, the remaining criteria — Relevance, Completeness and Actionability — were more notably influenced by the chosen configuration, both in the case of single-domain and multi-domain queries. For the sake of clarity, we discuss the result obtained for the latter criteria when considering two sample queries from Table 4 (a single-domain query, Q2, and a multi-domain query, Q1), whose quality scores are reported in Fig. 7. In the following, we examine the results obtained using C1 (No Ontology) and C5 (Full Ontology) configurations, thus comparing the baseline configuration with the configuration in which the FoCOSA ontology was applied throughout all RAG phases.

Across Single Selector runs (Fig. 5), C1 is not uniformly the worst: C2 (Ontology at Indexing only) yields the lowest Completeness (2.44) and Actionability (3.11), while C3 (Ontology at Retrieval only) attains the lowest Relevance (4.00) and Consistency (3.78). Nevertheless, C1 remains clearly suboptimal versus ontology-enhanced settings on most criteria.

Analysing the chat history of one of such multi-domain queries (specifically, Q1 in Appendix), it becomes evident that the Single Selector struggles to obtain relevant contextual data required for inference (in fact, answering Q1 necessitates data related to both the Consumer and Nutrition domains). In C1, even though a Multi Selector is employed for a multi-domain query such as Q1, the LLM lacks the contextual knowledge from the FoCOSA ontology modules to effectively relate Consumer and Nutrition data, resulting in low Relevance, Actionability and DUE scores.

In contrast, with ontology support, improvements appear at different stages: for Single Selector, C4 (Ontology at Response) yields the highest Relevance (4.89), Completeness (3.89) and Actionability (4.44), whereas C5 (Full Ontology) attains the best DUE (3.78) and remains top-2 on all other criteria (Fig. 5). Selector-wise (Fig. 6), under Full Ontology the Multi Selector experiment, improves Relevance (5.0 vs. 4.78), Consistency (5.0 vs. 4.67), and DUE (4.11 vs. 3.78); overall Actionability remains similar (4.33). At query level (Fig. 7), Q1 exhibits equal Actionability for both selectors (5.0), while DUE favors Multi (4.0 vs. 3.0). For Q2, Full-Ontology Single attains higher Actionability (5.0 vs. 4.0) with equal Relevance (5.0) and DUE (both 5.0).

This suggests that leveraging the ontology across all phases of RAG, as well as choosing the appropriate selector depending on whether the query is single-domain (Single Selector) or a multi-domain (Multi Selector), is an effective strategy to enhance response quality. At the same time, partial but strategically placed ontology use (C4 at Answer Generation) already lifts Single-Selector quality (best Relevance, Completeness, Actionability in Fig. 5), while end-to-end alignment (C5) maximises evidence usage (highest DUE) and remains competitive across all criteria.

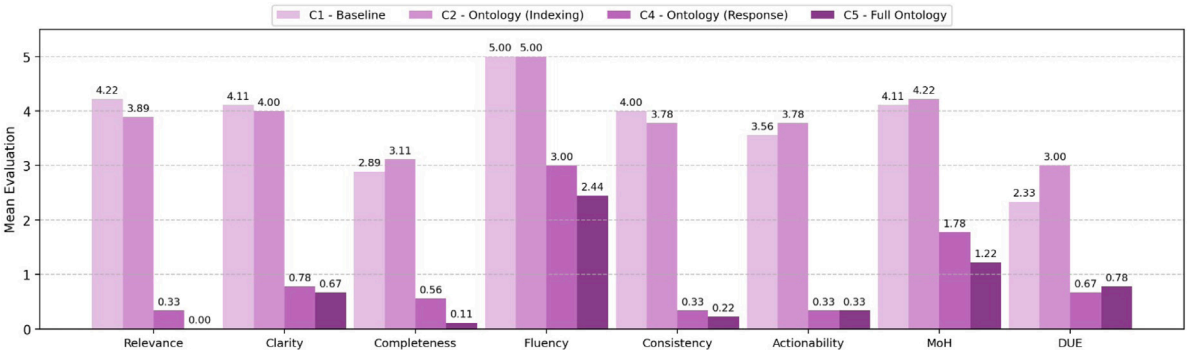


Fig. 8. Average quality criteria scores over configurations C1, C2, C4, and C5 using DeepSeek-R1 (14B) with Nomic embeddings.

Regardless of the adopted configuration, and focusing on context-related evaluation criteria, low Relevance and Actionability scores were tightly correlated with low DUE (Data Utilisation Effectiveness) values. A low DUE score implies that the response does not fully address the query, leading to diminished Relevance. Consequently, this also reduces the likelihood of generating practicable and feasible food advice, negatively affecting Actionability. Slight hallucination episodes were observed in Single Selector retrieval scenarios with multi-domain queries (particularly, when more than two domains have been involved). This is because, during inference, the LLM attempts to rely on its pre-trained knowledge when generating responses. As a result, the Single Selector approach limits access to comprehensive contextual data, leading to answers based on partial or insufficient information, deriving wrong or imprecise conclusions. This aligns with MoH patterns: for both Baseline and Full Ontology, Single shows higher MoH than Multi (Fig. 6), indicating greater hallucination risk when context is restricted. In Q1 under the Baseline setting, the high MoH (=5.0) stems from a refusal-style answer ("we do not have data") even though relevant evidence is present in the context; this failure to exploit available evidence directly explains the very low DUE (=1.0).

7.2.2. Impact of ontology utilisation on the RAG architecture

Figs. 5 and 6 show that not all partial ontology uses help: C2 (Indexing only) underperforms C1 on several criteria, and C3 (Retrieval only) yields low Relevance/Consistency. By contrast, C4 (Response only) delivers the strongest overall Single-Selector gains (best Relevance, Completeness, Actionability), while C5 (Full) provides the most robust balance with the highest DUE.

These results indicate that (i) indexing alone is insufficient if downstream stages are not ontology-aware; (ii) enforcing ontology constraints at answer generation (C4) can already lift quality substantially; and (iii) end-to-end ontology alignment (C5) maximises evidence usage (DUE) while remaining competitive across all criteria.

Overall, the experimental evaluation enabled to highlight the importance of ontology-based configurations, particularly when the FoCOSA ontology is used throughout all the phases. Multi Selector retrieval approaches further enhance performance when dealing with multi-domain queries. Instead, Single Selector methods remain prone to hallucinations in multi-domain queries. These insights will pave the way to further improve the RAG architecture, thus enhancing retrieval quality and response effectiveness. For instance, one of such improvements would be, considering the obtained medium-high Actionability results, ensuring that responses offer practical, implementable guidance in terms of food advice from the users' perspective, rather than just informational content. Concretely, we will refine the *Answer Generation* prompt (Fig. 4) with ontology-anchored, checklist-style constraints (conditions, contraindications, substitutions) to raise Actionability without sacrificing DUE.

7.2.3. Evaluation with open-source and lightweight language models

In order to further characterise the robustness of the proposed ontology-enhanced RAG architecture, we conducted an additional set of experiments using open-source and lightweight language models. In particular, we evaluated DeepSeek-R1 (14B) deployed via Ollama, combined with Nomic embeddings, across configurations C1 (No Ontology), C2 (Ontology at Indexing), C4 (Ontology at Response), and C5 (Full Ontology). Configuration C3 (Ontology at Retrieval) was excluded from this analysis due to the severe instability observed in preliminary tests when enabling LLM-based query rewriting, which consistently produced semantically misaligned or incoherent reformulations, regardless of whether ontology knowledge was injected or not.

The results reported in Fig. 8 show a systematic degradation of output quality when ontology knowledge is injected at response generation (C4) and across all RAG phases (C5). While C1 and C2 maintain acceptable levels of Relevance and Clarity, both metrics collapse under C4 and C5, indicating that the model fails to exploit the enriched semantic context and instead produces prompt-focused or structurally incoherent answers. Completeness exhibits a similar trend, peaking under C2 and sharply degrading under ontology-heavy configurations.

Interestingly, even Fluency, which is typically less sensitive to architectural variations, deteriorates under C4 and C5, decreasing from 5.00 (C1 and C2) to 3.00 (C4) and 2.44 (C5). This suggests that the semantic and structural complexity introduced by ontology-conditioned prompts exceeds the processing capacity of the model, leading not only to semantic errors but also to degraded linguistic quality.

Context-related criteria further confirm this behaviour. Data Utilisation Effectiveness (DUE) peaks under C2 and drops markedly under C4 and C5, indicating that retrieved evidence is not effectively exploited when ontology information is injected at later stages. In parallel, the low Minimization of Hallucinations (MoH) scores observed for ontology-heavy configurations reveal increased instances of irrelevant, incomplete, or prompt-focused answers. A qualitative inspection of the generated outputs shows that, in many cases, the model paraphrases prompt instructions or meta-level constraints instead of addressing the user query, a behaviour consistent with instruction collapse under high prompt complexity and with recent findings showing that lightweight and instruction-tuned models struggle to discriminate between competing instructions in long, structured prompts, often overfitting to surface patterns and latter prompt segments [60].

Additional experiments combining OpenAI embeddings with Gemma 3 (12B and 27B) models further confirmed that performance degradation persisted even when high-quality embeddings were retained. In particular, configurations C4 and C5 remained unstable, and query rewriting continued to produce unusable outputs, indicating that the observed limitations are not solely attributable to embedding quality, but primarily to the reasoning and instruction-following capacity of the generation model.

Overall, these results highlight a strong dependency between architectural expressiveness and model reasoning capacity. While ontology-enhanced RAG pipelines enable richer semantic grounding and multi-domain integration, they also impose a higher cognitive load on the generation model, which current lightweight and medium-scale open-source models are not yet able to sustain.

7.2.4. Threats to validity and implications for future work

After analysing the quality of LLM-generated responses, considering both configuration variations (Section 7.2.1) and the impact of the FoCOSA ontology utilisation on the RAG architecture (Section 7.2.2), we identified several factors that may threaten the validity of our findings and limit their broader applicability in real-world settings. These factors highlight certain gaps in the current framework; while they do not undermine the results, they point to concrete directions for further investigation and system evolution, which will be addressed in future work.

(a) Full ontology vs. modular employment of FoCOSA. Our current implementation of the RAG pipeline feeds the full ontology to the LLM in all the three phases of RAG. Nevertheless, this is a temporary choice intended to supply comprehensive domain knowledge in the LLM context during our preliminary experiments, and it is manageable given the limited size of the FoCOSA ontology. Looking ahead, we are working on an evolution of the framework to improve the current architecture by feeding the LLM only with the modules of FoCOSA ontology that are directly relevant to the user's query. This selective context would be determined dynamically based on the query whenever the FoCOSA ontology is used as input in the RAG pipeline, thus mitigating context window limitations and improving scalability.

(b) LLM flexibility vs. model–architecture compatibility. The RAG pipeline has been designed to be model-agnostic and supports the invocation of both proprietary LLMs (e.g., Gemini, Claude) and open-source models (e.g., via the Ollama framework). However, the experimental analysis highlights that the effectiveness of the proposed ontology-enhanced architecture is strongly conditioned by the reasoning capacity and instruction-following robustness of the underlying model.

In particular, architectures that rely on multi-phase semantic integration, ontology-conditioned prompting, and multi-domain reasoning impose a cognitive load that exceeds the practical capabilities of current lightweight models. This introduces a validity boundary: while the architectural design is general, its full benefits can only be realised by models that are able to discriminate between multiple instruction sources, integrate structured domain knowledge, and sustain long-context reasoning.

This dependency does not undermine the generality of the approach, but rather delineates its current operational envelope. It also has direct implications for deployment: practitioners adopting ontology-enhanced RAG pipelines should carefully consider model capacity as a first-class design parameter. As open-source models continue to evolve in terms of instruction discrimination, reasoning depth, and long-context handling, a key direction for future work is to systematically assess whether and under which conditions the proposed architectural approach can be effectively supported by such models.

(c) Impact of FoCOSA evolution on RAG outputs. In case of evolutions to the FoCOSA ontology, the updated version of FoCOSA would be used as input across all three RAG phases components (query preprocessing, chunks description generation, and answer generation), ensuring consistency with the domain latest knowledge. Amongst these three components, the chunks description generation component may have a relatively higher impact upon changes in FoCOSA semantic structure, with respect to the others. Indeed, any evolution of the ontology (such as class renaming, refinement of definitions, or the addition, removal, or reorganisation of concepts/relationships and modules) may affect the alignment between previously generated chunk summaries and the current ontology version. Since these summaries are produced during the indexing phase and used as descriptive metadata for chunk embeddings, such changes may lead to discrepancies in terminology or conceptual scope with respect to the updated ontology. In the experimental evaluation presented in Section 7, we assumed a fixed ontology version across all phases of the RAG pipeline; however, ontology evolution represents an open challenge for long-running or production deployments. In future evolutions of the system, this issue can be addressed by adopting established ontology engineering practices, including the following complementary strategies: (i) associating each chunk not only with a textual summary but also with stable entity identifiers (URIs) and explicit ontology version metadata, so that the system relies on consistent identifiers rather than potentially changing labels or textual names; (ii) performing change-impact analysis on ontology updates to selectively re-generate summaries and embeddings only for

chunks that reference affected entities, avoiding the need to re-index the entire corpus; and (iii) exploiting ontology changelogs and explicit mappings (e.g., old-to-new class names or equivalent entities) at query rewriting or retrieval time to maintain backward compatibility; when these mappings are insufficient, summaries and embeddings can be regenerated on demand and cached, so that frequently accessed content is updated efficiently without repeating the same computation. The modular structure of FoCOSA and the explicit separation between ontology management, summary generation, and vector storage provide a solid foundation for integrating these mechanisms as the system evolves, facilitating systematic evaluation of how different ontology changes impact the RAG pipeline. Consequently, as part of future work, we will perform simulation-based experiments to quantify the pipeline sensitivity to different classes of ontology changes, and to derive principled criteria and trade-offs for deciding when chunk summaries and embeddings should be updated selectively (only for affected chunks) or globally (by re-indexing the entire corpus).

8. Concluding remarks

In this paper, we introduced an ontology-enhanced conversational food advisory system that integrates a modular ontology within a Retrieval-Augmented Generation (RAG) framework to improve the contextual awareness and accuracy of food recommendations provided by Large Language Models (LLMs).

Our contribution lies in integrating a modular ontology with LLMs, in three distinct stages of the approach including the embedding description generation, the query processing and rewriting, and the response generation. By systematically combining LLMs' natural language understanding with ontological structured representation of knowledge, our method addresses the challenges of recommendation oversimplification, over-personalisation, and knowledge inconsistencies.

Experimental results confirm that ontology-enhanced RAG-based LLMs improve recommendation accuracy, reduce hallucinations, and better align responses with nutritional, sustainability, and lifestyle constraints. These findings underscore the benefits of structured knowledge integration in LLM-driven food advisory systems. Future research will explore more advanced RAG techniques for optimising retrieval accuracy and integrating structured and unstructured data. Additionally, large-scale user studies will help validate our approach using standard evaluation metrics. The FoCOSA ontology will be continuously refined through user feedback and expanded to incorporate emerging dietary trends and further sustainability aspects. Moreover, addressing the threats to validity and the open gaps identified in Section 7.2.4 will play a key role in guiding these future developments, contributing to the robustness, scalability, and long-term applicability of the proposed system.

CRediT authorship contribution statement

Ada Bagozi: Writing – review & editing, Writing – original draft, Software, Data curation, Conceptualization. **Devis Bianchini:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization. **Massimiliano Garda:** Writing – review & editing, Software, Data curation. **Michele Melchiori:** Writing – review & editing, Writing – original draft, Conceptualization. **Anisa Rula:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Data curation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Anisa Rula reports financial support was provided by Italian Ministry of University and Research (MIUR). If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Examples of query answering and output quality evaluation

Query 1 (No Ontology - SingleSourceSelector)

Query: How does salmon impact the health of customers with cardiovascular conditions?
Response: *The information provided does not include specific health effects of salmon on individuals with cardiovascular conditions. Therefore, the data is not enough to fully answer the query.*
[...]
DUE score: 1
Relevance score: 2
Relevance justification: *The response provided acknowledges the inadequacy of information about the specific issue being questioned. Instead of trying to provide a potentially false or misleading answer, the response properly mentions the lack of sufficient data to answer the question. As this response accurately echoes the query's focus on salmon's impact on individuals with cardiovascular conditions, it's considered to be relevant, although lacking in detailed information.*
Completeness score: 0
Completeness justification: *The response does not provide any information to answer the query about the impact of salmon on the health of customers with cardiovascular conditions. It does not mention the nutrients present in salmon, such as omega-3 fatty acids, which are known to be beneficial for cardiovascular health, nor does it discuss any potential risks. Therefore, the answer is not complete, as it lacks essential information to the query.*
Actionability score: 0
Actionability justification: *The response indeed contains accurate statement, people can't act upon this because it does not provide any concrete information as to how salmon impacts cardiovascular health. This answer does not provide any action steps, guidance, or recommendations about consuming salmon for individuals with cardiovascular conditions. Thus, it performs poorly on the actionability metric.*

Query 1 (No Ontology - MultiSourceSelector)

Query: How does salmon impact the health of customers with cardiovascular conditions?
Response: *The information provided does not include specific health effects of salmon on individuals with cardiovascular conditions. Therefore, the data is not enough to fully answer the query regarding the health effects of salmon on such individuals.*
[...]
DUE score: 1
Relevance score: 0
Relevance justification: *The response provided avoids answering the direct question about the impacts of consuming salmon on people with cardiovascular conditions. Instead, it informs about an inability to provide relevant information due to lack of it, which doesn't contribute any value to the query. The task was on evaluating the health effects of salmon, however the respondent does not provide any substantive answer, thus failing the relevance metric.*
Completeness score: 0
Completeness justification: *The response to the query does not provide the desired information. The question asks for the health impact of salmon on individuals with cardiovascular conditions. In response, the individual acknowledges that they do not provide the required information, which does not address the question at all. There's no mention of salmon's nutritional content—such as omega-3 fatty acids, which are known to be heart-healthy—or any potential effects, positive or negative, for people with cardiovascular conditions. Hence, it lacks completion.*
Actionability score: 0
Actionability justification: *The response does not offer any action points nor does it directly answer the query. While it honestly states the lack of specific information about the health effects of salmon on individuals with cardiovascular conditions, it doesn't provide any further actionable advice, e.g., suggesting that the individual consult with a healthcare professional, or recommend further research. The recipient of this message does not have any material to act on or any clearer understanding of the issue after reading this response.*

Query 1 (Full Ontology - SingleSourceSelector)

Query: How does salmon impact the health of customers with cardiovascular conditions?
Response: *The impact of salmon on consumers with cardiovascular health conditions can be assessed by examining the nutritional profile of salmon and the dietary restrictions of individuals with such conditions. Salmon is known for its high omega-3 fatty acid content, which is beneficial for heart health. Omega-3s can help reduce inflammation, lower blood pressure, and decrease triglycerides, all of which are important for individuals with cardiovascular issues. Additionally, salmon is a good source of protein and contains essential vitamins and minerals, such as vitamin D and selenium, which can support overall health.*
However, consumers with specific dietary restrictions due to cardiovascular health conditions may need to consider how salmon fits into their diets. For example:
1. Catherine Gomez has health conditions including hypertension and heart disease, and follows a strict diet that is keto, dairy-free, vegetarian, gluten-free, and nut-free. While salmon is generally heart-healthy, it may not align with her vegetarian preference.
2. Sara Rogers also has heart disease and diabetes, and follows a vegetarian and gluten-free diet. Similar to Catherine, salmon would not be suitable for her due to her vegetarian restriction.
3. Kelly Clark has diabetes and heart disease, and follows a vegetarian, dairy-free, and nut-free diet. Again, salmon would not be appropriate for her dietary choices.
4. Patrick Stewart has heart disease and diabetes, and follows a vegetarian, keto, and nut-free diet. Salmon would not fit into his vegetarian diet.
5. Elizabeth Gonzalez has diabetes and heart disease, and follows a gluten-free diet. While salmon is gluten-free, it may not be a preferred choice for her if she is focusing on plant-based options.
In summary, while salmon can be beneficial for heart health due to its omega-3 fatty acids, its suitability for consumers with cardiovascular health conditions largely depends on their specific dietary restrictions. For those who follow vegetarian or vegan diets, alternative sources of omega-3s, such as flaxseeds or chia seeds, may be more appropriate.
[...]
DUE score: 5
Relevance score: 5
Relevance justification: *The response is highly relevant to the query. It firstly explains the benefits of salmon, especially its high omega-3 content, for individuals with heart conditions. The response then cleverly details the impact of salmon on various individuals with cardiovascular issues but who also follow certain dietary restrictions. Illustrative use of personal cases connects the dietary assessment with possible real world scenarios. The focus is precisely on the impact of salmon, satisfying the original query to great extent. There might be need for more specific cases unique to cardiovascular conditions which is normal, but these are dictated by the specifications in the query, which here are essentially broad.*
Completeness score: 4
Completeness justification: *The response is comprehensive; it describes the nutritional profile of salmon, explaining how each characteristic impacts cardiovascular health. The valuable addition was discussing how the nutritional benefits of salmon could be negated by a consumer's dietary preferences or restrictions, thereby realizing that individual diets can vastly differ. The response illustrates this by providing hypothetical examples of individuals with different dietary restrictions and details why salmon wouldn't technically suit their diets despite its heart-health benefits. Notably, the response also offers an alternative (flaxseeds and chia seeds) for those who, due to diet restrictions, cannot consume salmon. Besides, each underlying health condition isn't directly addressed in terms of the benefits or drawbacks of consuming salmon. The reply is mostly centred on dietary restrictions. However, overall, the explanation is comprehensive and informative while directly addressing the query.*
Actionability score: 4
Actionability justification: *The response provides an excellent, detailed, and informative overview of the impact of salmon on cardiovascular health. It mentions the health benefits of salmon and why it could be good for people with heart-related conditions. The response then mentions a key factor—dietary restrictions—that could influence its suitability for an individual's diet. Essentially, it recognizes that while salmon has cardiovascular benefits, it might not be suitable for everyone - and even suggests alternatives. The examples given provide practical instances where these factors can influence decision-making. However, the response could offer more direct actionable steps or simpler guidelines for consumers who are trying to incorporate healthier options into their diets. Some depth of information relevant for professionals is less directly applicable for consumers seeking immediate next steps.*

Query 1 (Full Ontology - MultiSourceSelector)

Query: How does salmon impact the health of customers with cardiovascular conditions?
Response: *The health impact of salmon on consumers with cardiovascular health conditions can be quite positive, particularly due to its nutritional profile. Salmon is rich in omega-3 fatty acids, which are known to support heart health by reducing inflammation, lowering blood pressure, and decreasing triglyceride levels. These benefits are particularly relevant for individuals with cardiovascular conditions.*
For consumers like Sara Rogers and Kelly Clark, who have health conditions such as diabetes and heart disease, incorporating salmon into their diet can be beneficial. Salmon is also a good source of high-quality protein, vitamins (such as B12 and D), and minerals (like selenium), which contribute to overall health. However, it is essential for these consumers to consider their dietary restrictions. For instance, Sara Rogers follows a vegetarian and gluten-free diet, which would exclude salmon from her options. On the other hand, Kelly Clark, who is vegetarian but also has dietary restrictions related to nuts and dairy, may need to find alternative sources of omega-3s that align with her vegetarian lifestyle.
In summary, for those who can include salmon in their diet, it offers significant cardiovascular benefits. However, individuals with specific dietary restrictions must seek alternative sources of similar nutrients to maintain their health.
[...]
DUE score: 5
Relevance score: 5
Relevance justification: *The response is highly relevant to the query, providing an in-depth explanation on the potential benefits of salmon for those with cardiovascular health conditions. This includes details about specific nutrients that contribute to heart health and how they could influence well-being. The answer even considers the dietary restrictions of named individuals and discusses options for those who can't consume salmon, displaying a broad understanding of the topic. In terms of relevance, it does an excellent job targeting the main question, thus the high score.*
Completeness score: 4
Completeness justification: *The provided response is thorough and well-detailed, providing valuable insights on the health benefits of salmon, especially for those with cardiovascular conditions. It highlights the nutritional content of salmon, such as omega-3 fatty acids, high-quality protein, vitamins, and minerals, and how they support heart health.*
The response also considers individual dietary restrictions, demonstrating a comprehensive understanding of nutritional needs and personal diet preferences. The individual cases of Sara Rogers and Kelly Clark provide real-world relevance to the theoretical nutritional discussion. However, the question explicitly asked for the impact on "customers with cardiovascular conditions", but the instances used involve people with additional other health conditions and dietary restrictions, which might not be wholly applicable to the query asked.
Overall, this response does an admirable job in connecting nutritional facts about salmon with their impact on cardiovascular health while acknowledging individual dietary differences.
Actionability score: 4
Actionability justification: *The response provides good insight on how salmon can affect the health of customers with cardiovascular conditions and why it is beneficial due to its rich content in such as omega-3 fatty acids, high-quality protein, vitamins, and minerals. The response consider different approaches by introducing the cases of two individuals (Sara Rogers and Kelly Clark) who have different dietary restrictions, which is an essential aspect in terms of actionability. The information regarding dietary restrictions, potential alternatives and considerations for diverse diets also enhances the actionability of the response. In general, it shows a good degree of actionability even though it is not entirely prescriptive in its advice.*

Query 2 (No Ontology - SingleSourceSelector)

Query: What macronutrients and micronutrients are present in salmon?
Response: *The information provided does not include specific nutritional data for salmon. Therefore, I cannot list the macronutrients and micronutrients found in salmon based on the available data.*
[...]
DUE score: 1
Relevance score: 1
Relevance justification: *The response is related to the query as it addresses the topic of nutrients in salmon. However, it provides no actual information about the macronutrients and micronutrients present in salmon, which is what the query is asking for. Although it stays on topic, it does not answer the query or provide any relevant detail, making it largely useless in terms of the user's need for information.*
Completeness score: 0
Completeness justification: *The response given does not contain any of the information requested in the query. The query asks for an explanation of the macronutrients and micronutrients in salmon. An adequate answer would need to list out these nutrients and possibly also provide some details about their impact on health or dietary importance. The respondent hasn't given any such information, not even a single nutrient name, and only expressed their inability to provide the information. There's no completeness in this response as it lacks any actual content related to the query.*
Actionability score: 0
Actionability justification: *The response falls short on the actionability metric. The user asked for specific information about the macronutrients and micronutrients present in salmon, and the response failed to provide this information. It does not guide the user towards finding the desired information or offer any actionable steps on how the user can obtain this knowledge. The answer does not contribute to the initial query, leaving the user without any further guidance or actionable advice.*

Query 2 (Full Ontology - SingleSourceSelector)

Query: What macronutrients and micronutrients are present in salmon?
Response: *The nutritional components of the food item "Salmon" (specifically "Fish, bones removed (Alaska Native), canned, red, salmon") per 100 grams are as follows:*

Macronutrients:
- Calories: 161 kcal
- Total Fat: 5.8 g
- Saturated Fat: 0.9 g
- Cholesterol: 69 mg
- Sodium: 390 mg
- Protein: 27.31 g
- Carbohydrates: 5.76 g
- Fibre: 0 g
- Sugars: 0 g

Vitamins and Minerals:
- Vitamin A: 216 IU (65 mcg)
- Vitamin B12: 4.91 mcg
- Vitamin B6: 0.103 mg
- Vitamin C: 0 mg
- Vitamin D: 0 mcg
- Vitamin E: 0.61 mg
- Vitamin K: 0.0 mcg
- Calcium: 28 mg
- Iron: 1.90 mg
- Magnesium: 32 mg
- Phosphorus: 260 mg
- Potassium: 370 mg
- Selenium: 40.3 mcg
- Zinc: 0.66 mg

This comprehensive breakdown provides insight into the nutritional value of canned red salmon, highlighting its protein content and various vitamins and minerals beneficial for health.

[...]
DUE score: 5
Relevance score: 5
Relevance justification: *The response accurately and thoroughly provides the requested details about both macronutrients and micronutrients present in salmon. It covers information on calories, fats, cholesterol, sodium, protein, carbohydrates, and a large range of vitamins and minerals, as per the query. Furthermore, it even details out the subcomponents of fats and carbohydrates. The response underlines the specific type of salmon in discussion, i.e. "canned red salmon", providing precise and contextual information. Hence, the information provided is relevant and fitting to the query.*
Completeness score: 5
Completeness justification: *The response given is thorough and complete. It lists out both the macronutrients (Calories, Total Fat, Saturated Fat, Cholesterol, Sodium, Protein, Carbohydrates, Fibre, Sugar) and micronutrients (various vitamins and minerals) found in the specified type of salmon. The quantities of each nutrient per 100 grams of salmon are also provided for more precise information. Notably, it includes both the more common nutrients one might expect, such as protein and vitamins, and less frequently mentioned nutrients, like Selenium and Phosphorus. This response thoroughly satisfies the requirements of the query about what macronutrients and micronutrients are present in salmon.*
Actionability score: 2
Actionability justification: *The response provided is informative and detailed, outlining the nutritional breakdown of salmon per 100 grams. It covers macronutrients including calories, fat, cholesterol, sodium, protein, and carbohydrates, as well as several vitamins and minerals. However, for the metric of actionability – the capacity to use this information to make a decision or take an action – the response falls short. The information provided is crucial but lacks suggestions on how to integrate salmon into one's diet, its health benefits, possible drawbacks, serving suggestions or alternatives for those who may have dietary restrictions or allergies. Because the metric is actionability, which anticipates that the response will guide the user in how to use this information in making dietary decisions, a lower score is justified.*

Query 2 (Full Ontology - MultiSourceSelector)

Query: What macronutrients and micronutrients are present in salmon?
Response: *The nutritional information for the food item "salmon" includes a comprehensive breakdown of macronutrients and micronutrients. Here are the details based on various preparations of salmon:*
1. Calories: Ranges from 92 to 640 kcal per 100 g depending on the preparation method.
2. Total Fat: Varies significantly, with values ranging from 0.5 g to 81 g.
- Saturated Fat: Typically around 0.2 g to 36 g.
- Monounsaturated Fat: Present in varying amounts depending on the preparation.
- Polyunsaturated Fat: Also varies based on cooking method.
3. Cholesterol: Ranges from 23 mg to 144 mg.
4. Sodium: Can range from 23 mg to 1008 mg.
5. Protein: Generally high, ranging from 17 g to 28 g.
6. Carbohydrates: Usually low, often around 0 g to 2 g.
- Fibre: Typically negligible.
- Sugars: Generally low or absent.

Vitamins and Minerals:
- Vitamin A: Varies, with some preparations providing significant amounts (up to 496 IU).
- Vitamin B12: Present in substantial amounts, often exceeding 2.0 mcg.
- Vitamin D: Found in varying quantities, often around 0.0 to 10.0 mcg.
- Selenium: Typically present, contributing to the overall nutritional profile.
- Calcium, Iron, Magnesium, and Potassium: Present in varying amounts, contributing to the mineral content.

Specific Preparations:
- Canned Salmon: Generally lower in fat and calories compared to fried or coated versions.
- Fried Salmon: Higher in calories and fat due to the cooking method.
- Steamed or Poached Salmon: Typically retains more nutrients with lower fat content.

This nutritional information highlights salmon as a rich source of protein and essential fatty acids, making it a valuable addition to a balanced diet, particularly for those who do not have dietary restrictions against fish.

[...]
DUE score: 4
Relevance score: 5
Relevance justification: *The response is directly relevant to the query. It provides an exhaustive breakdown of the macronutrients and micronutrients present in salmon, as well as mentions of variations based on different preparation methods, which aligns with the question's requirements. The response demonstrates a clear focus on nutritional metrics associated with salmon, making it highly relevant to the question asked.*
Completeness score: 5
Completeness justification: *The response is outstanding in the context of the asked question, and provides a comprehensive list of both macronutrients and micronutrients found in salmon. The information is well-organized and includes ranges for the different types of preparations of salmon. The response also breaks down both polyunsaturated and monounsaturated fats, instead of only accounting for total fat content. On top of that, it provides details about specific types of preparation for salmon, offering added context. Lastly, the response also goes beyond the original query to point out the potential health benefits of consuming salmon, which is valuable information for the user.*
Actionability score: 3
Actionability justification: *The response provided a comprehensive breakdown of the macronutrients and micronutrients found in salmon, their ranges depending on the preparation, and even gave some information about different preparation methods. However, while the answer is very informative, it falls somewhat short on the actionability metric. This is because, although the response suggests that salmon is a good source of protein and essential fatty acids, it does not provide clear recommendations on how to incorporate salmon into one's diet, what preparation methods would be best for specific nutritional needs, or how frequently it should be consumed.*

Data availability

We use a repository for sharing our data.

References

[1] G. Sedrakyan, A. Gavai, J. van Hillegersberg, Design implications towards human-centric semantic recommenders for sustainable food consumption, in: T.P. Sales, J. Araújo, J. Borbinha, G. Guizzardi (Eds.), Advances in Conceptual Modeling - ER 2023 Workshops, CMLS, CMOMM4FAIR, EmpER, JUSMOD, OntoCom, QUAMES, and SmartFood, Lisbon, Portugal, November 6-9, 2023, Proceedings, in: Lecture Notes in Computer Science, vol. 14319, Springer, 2023, pp. 312–328.

[2] S. Haussmann, O. Seneviratne, Y. Chen, Y. Ne'eman, J.V. Codella, C. Chen, D.L. McGuinness, M.J. Zaki, FoodKG: A semantics-driven knowledge graph for food recommendation, in: C. Ghidini, O. Hartig, M. Maleshkova, V. Svátek, I.F. Cruz, A. Hogan, J. Song, M. Lefrançois, F. Gandon (Eds.), The Semantic Web - ISWC 2019 - 18th International Semantic Web Conference, Auckland, New Zealand, October 26-30, 2019, Proceedings, Part II, in: Lecture Notes in Computer Science, vol. 11779, Springer, 2019, pp. 146–162.

[3] G. Simsek-Senel, H. Rijgersberg, B. Öztürk, J. Weits, A. Fensel, I-KNOW-FOO: Interlinking and creating knowledge graphs for near-zero CO2 emission diets and sustainable FOOd production, in: AI, Data, and Digitalization, 2024, pp. 106–119.

[4] A.B. Rashid, M.A.K. Kausik, AI revolutionizing industries worldwide: A comprehensive overview of its diverse applications, Hybrid Adv. 7 (2024) 100277.

[5] A.D. Starke, C. Musto, A. Rapp, G. Semeraro, C. Trattner, "Tell me why": using natural language justifications in a recipe recommender system to support healthier food choices, User Model. User Adapt. Interact. 34 (2) (2024) 407–440.

[6] J. Li, T. Tang, W.X. Zhao, J. Nie, J. Wen, Pre-trained language models for text generation: A survey, ACM Comput. Surv. 56 (9) (2024) 230:1–230:39.

[7] A. Rostami, An Integrated Framework for Contextual Personalized LLM-Based Food Recommendation (Ph.D. thesis), University of California, Irvine, 2024, URL: https://escholarship.org/.

[8] Z. Zhao, W. Fan, J. Li, Y. Liu, X. Mei, Y. Wang, Z. Wen, F. Wang, X. Zhao, J. Tang, Q. Li, Recommender systems in the era of large language models (LLMs), IEEE Trans. Knowl. Data Eng. 36 (11) (2024) 6889–6907.

[9] Y. Gao, Y. Xiong, X. Gao, K. Jia, J. Pan, Y. Bi, Y. Dai, J. Sun, Q. Guo, M. Wang, H. Wang, Retrieval-augmented generation for large language models: A survey, 2023, CoRR abs/2312.10997.

[10] X. Wang, Z. Wang, X. Gao, F. Zhang, Y. Wu, Z. Xu, T. Shi, Z. Wang, S. Li, Q. Qian, R. Yin, C. Lv, X. Zheng, X. Huang, Searching for best practices in retrieval-augmented generation, 2024, arXiv:2407.01219. URL: https://arxiv.org/abs/2407.01219.

[11] T. Baldazzi, L. Bellomarini, S. Ceri, A. Colombo, A. Gentili, E. Sallinger, Fine-tuning large enterprise language models via ontological reasoning, in: A. Fensel, A. Ozaki, D. Roman, A. Soylu (Eds.), Rules and Reasoning - 7th International Joint Conference, RuleML+RR 2023, Oslo, Norway, September 18-20, 2023, Proceedings, in: Lecture Notes in Computer Science, vol. 14244, Springer, 2023, pp. 86–94.

[12] F. Ronzano, J. Nanavati, Towards ontology-enhanced representation learning for large language models, 2024, arXiv:2405.20527. URL: https://arxiv.org/abs/2405.20527.

[13] D. Allemang, J. Sequeda, Increasing the LLM accuracy for question answering: Ontologies to the rescue!, 2024, CoRR.

[14] S. Es, J. James, L.E. Anke, S. Schockaert, Ragas: Automated evaluation of retrieval augmented generation, in: Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations, 2024, pp. 150–158.

[15] L. Zheng, W.-L. Chiang, Y. Sheng, S. Zhuang, Z. Wu, Y. Zhuang, Z. Lin, Z. Li, D. Li, E.P. Xing, H. Zhang, J.E. Gonzalez, I. Stoica, Judging LLM-as-a-judge with MT-bench and chatbot arena, 2023, arXiv:2306.05685. URL: https://arxiv.org/abs/2306.05685.

[16] J. Amidei, P. Piwek, A. Willis, The use of rating and Likert scales in natural language generation human evaluation tasks: A review and some recommendations, in: K. van Deemter, C. Lin, H. Takamura (Eds.), Proceedings of the 12th International Conference on Natural Language Generation, 2019, pp. 397–402.

[17] Z. Chen, Y. Liu, L. Chen, S. Zhu, M. Wu, K. Yu, Opal: ontology-aware pretrained language model for end-to-end task-oriented dialogue, Trans. Assoc. Comput. Linguist. 11 (2023) 68–84.

[18] C. Aceta, I. Fernández, A. Soroa, Todo: A core ontology for task-oriented dialogue systems in industry 4.0, in: Further with Knowledge Graphs, 2021, pp. 1–15.

[19] X. Dong, J. Chen, PluDG: enhancing task-oriented dialogue system with knowledge graph plug-in module, PeerJ Comput. Sci. 9 (2023) e1707.

[20] V.I. Iga, G.C. Silaghi, Ontology-based dialogue system for domain-specific knowledge acquisition, in: Information Systems Development, Organizational Aspects and Societal Trends, ISD2023, 2023.

[21] K. Sharma, P. Kumar, Y. Li, OG-RAG: Ontology-grounded retrieval-augmented generation for large language models, 2024, arXiv:2412.15235. URL: https://arxiv.org/abs/2412.15235.

[22] T.C. Nemtoc, A.-M. Ghiran, Natural language querying of invoice data using RAG and graphrag: Leveraging LLMs for financial document insights, in: International Conference on Advanced Information Systems Engineering, 2025, pp. 69–80.

[23] B. Peng, Y. Zhu, Y. Liu, X. Bo, H. Shi, C. Hong, Y. Zhang, S. Tang, Graph retrieval-augmented generation: A survey, 2024, arXiv preprint arXiv:2408.08921.

[24] P. Sarthi, S. Abdullah, A. Tuli, S. Khanna, A. Goldie, C.D. Manning, Raptor: Recursive abstractive processing for tree-organized retrieval, in: The Twelfth International Conference on Learning Representations, 2024.

[25] D. Edge, H. Trinh, N. Cheng, J. Bradley, A. Chao, A. Mody, S. Truitt, D. Metropolitansky, R.O. Ness, J. Larson, From local to global: A graph rag approach to query-focused summarization, 2024, arXiv preprint arXiv:2404.16130.

[26] S. Ma, C. Xu, X. Jiang, M. Li, H. Qu, C. Yang, J. Mao, J. Guo, Think-on-graph 2.0: Deep and faithful large language model reasoning with knowledge-guided retrieval augmented generation, in: The Thirteenth International Conference on Learning Representations, 2025.

[27] H. Liu, S. Wang, Y. Zhu, Y. Dong, J. Li, Knowledge graph-enhanced large language models via path selection, in: Findings of the Association for Computational Linguistics ACL 2024, 2024, pp. 6311–6321.

[28] Y. Wang, N. Lipka, R.A. Rossi, A. Siu, R. Zhang, T. Derr, Knowledge graph prompting for multi-document question answering, in: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 38, 2024, pp. 19206–19214, 17.

[29] D.M.R. Bara, D.M. Mesesan, G.C. Silaghi, Retrieval-augmented generation for entity alignment in knowledge graphs: An incipient experiment, in: International Conference on Advanced Information Systems Engineering, 2025, pp. 50–62.

[30] K. Cheng, N.K. Ahmed, T. Willke, Y. Sun, Structure guided prompt: Instructing large language model in multi-step reasoning by exploring graph structure of the text, 2024, arXiv preprint arXiv:2402.13415.

[31] R. Yang, B. Yang, X. Zhao, F. Gao, A. Feng, S. Ouyang, M. Blum, T. She, Y. Jiang, F. Lecue, et al., Graphusion: A RAG framework for scientific knowledge graph construction with a global perspective, in: Companion Proceedings of the ACM on Web Conference 2025, 2025, pp. 2579–2588.

[32] Z. Xu, M.J. Cruz, M. Guevara, T. Wang, M. Deshpande, X. Wang, Z. Li, Retrieval-augmented generation with knowledge graphs for customer service question answering, in: Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval, 2024, pp. 2905–2909.

[33] D. Henselmann, R. Dorsch, A. Harth, Impact of knowledge graph representations on question answering with language models, in: International Conference on Advanced Information Systems Engineering, 2025, pp. 81–92.

[34] D.M. Dooley, L. Andrés-Hernández, G. Bordea, L. Carmody, D. Cavalieri, L.E. Chan, P. Castellano-Escuder, C. Lachat, F. Mougin, F. Vitali, C. Yang, M. Weber, H. Küçük-McGinty, M. Lange, OBO foundry food ontology interconnectivity, Semant. Web 15 (4) (2024) 1239–1258.

[35] D.M. Dooley, E.J. Griffiths, G.S. Gosal, P.L. Buttigieg, R. Hoehndorf, M.C. Lange, L.M. Schriml, F.S. Brinkman, W.W. Hsiao, Foodon: a harmonized food ontology to increase global food traceability, quality control and data integration, NPJ Sci. Food 2 (1) (2018) 23.

[36] Z. Qi, Y. Yu, M. Tu, J. Tan, Y. Huang, FoodGPT: A large language model in food testing domain with incremental pre-training and knowledge graph prompt, 2023, arXiv:2308.10173. URL: https://arxiv.org/abs/2308.10173.

[37] P. Zhou, W. Min, C. Fu, Y. Jin, M. Huang, X. Li, S. Mei, S. Jiang, FoodSky: A Food-oriented Large Language Model that Passes the Chef and Dietetic Examination, 2024, arXiv preprint arXiv:2406.10261.

[38] B.U. Rani, T. Joshnavalli, B.S. Reddy, A. Sreelaasya, An advanced deep learning approach for dietary recommendations using ROBERTA, in: 2023 4th International Conference on Intelligent Technologies, CONIT, IEEE, 2024, pp. 1–6.

[39] I. Papastratis, D. Konstantinidis, P. Daras, K. Dimitropoulos, AI nutrition recommendation using a deep generative model and ChatGPT, Sci. Rep. 14 (1) (2024) 14620.

[40] F. Yang, Z. Chen, Z. Jiang, E. Cho, X. Huang, Y. Lu, PALR: Personalization aware LLMs for recommendation, 2023, arXiv:2305.07622. URL: https://arxiv.org/abs/2305.07622.

[41] G. Zhang, User-centric conversational recommendation: Adapting the need of user with large language models, in: Proceedings of the 17th ACM Conference on Recommender Systems - RecSys '23, ACM, 2023, pp. 1349–1354.

[42] B. Hannon, Y. Kumar, J.J. Li, P. Morreale, Chef dalle: Transforming cooking with multi-model multimodal AI, Comput. 13 (7) (2024) 156.

[43] P. Kansaksiri, P. Panomkhet, N. Tantisuwichwong, Smart cuisine: Generative recipe & chatgpt powered nutrition assistance for sustainable cooking, Procedia Comput. Sci. 225 (2023) 2028–2036.

[44] Z. Zhang, Z. Wang, T. Ma, V.S. Taneja, S. Nelson, N.H.L. Le, K. Murugesan, M. Ju, N.V. Chawla, C. Zhang, et al., MOPI-HFRS: A multi-objective personalized health-aware food recommendation system with LLM-enhanced interpretation, 2024, arXiv preprint arXiv:2412.08847.

[45] I. Dergaa, H.B. Saad, H. Ghouili, J.M. Glenn, A. El Omri, I. Slim, Y. Hasni, M. Taheri, M.B. Aissa, N. Guelmami, et al., Evaluating the applicability and appropriateness of ChatGPT as a source for tailored nutrition advice: A multi-scenario study, New Asian J. Med. 2 (1) (2024) 1–16.

[46] I. Papastratis, A. Stergioulas, D. Konstantinidis, P. Daras, K. Dimitropoulos, Can ChatGPT provide appropriate meal plans for NCD patients? Nutrition 121 (2024) 112291.

[47] Y.S. Soekamto, A. Lim, L.C. Limanjaya, Y.K. Purwanto, S.-H. Lee, D.-K. Kang, Pic2Plate: A vision-language and retrieval-augmented framework for personalized recipe recommendations, Sens. 25 (2) (2025) 449.

[48] G. Liu, H. Yin, B. Zhu, J. Chen, C.-W. Ngo, Y.-G. Jiang, Retrieval augmented recipe generation, in: 2025 IEEE/CVF Winter Conference on Applications of Computer Vision, WACV, 2025, pp. 2453–2463.

[49] K. Kalpakoglou, L.C. Pérez, N. Boqué, Ç. Erdoğan, M. Guldas, L.P. Gymnopoulos, K. Dimitropoulos, An AI-based nutrition recommendation system: technical validation with insights from mediterranean cuisine, Front. Nutr. 12 (2025) 1546107.

[50] Y. Mao, P. He, X. Liu, Y. Shen, J. Gao, J. Han, W. Chen, Generation-augmented retrieval for open-domain question answering, in: Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers), 2021, pp. 4089–4100.

[51] A. Bagozi, D. Bianchini, M. Melchiori, A. Rula, Enhancing LLMs contextual knowledge with ontologies for personalised food recommendation, in: M. Barhamgi, H. Wang, X. Wang (Eds.), Web Information Systems Engineering – WISE 2024, Springer Nature Singapore, Singapore, 2025, pp. 273–283.

[52] M. Poveda-Villalón, A. Fernández-Izquierdo, M. Fernández-López, R. García-Castro, LOT: An industrial oriented ontology engineering framework, Eng. Appl. Artif. Intell. 111 (2022) 104755.

[53] C.M. Keet, Z.C. Khan, Discerning and characterising types of competency questions for ontologies, 2024, arXiv:2412.13688. URL: https://arxiv.org/abs/2412.13688.

[54] Y. Gao, Y. Xiong, X. Gao, K. Jia, J. Pan, Y. Bi, Y. Dai, J. Sun, M. Wang, H. Wang, Retrieval-augmented generation for large language models: A survey, 2024, arXiv:2312.10997. URL: https://arxiv.org/abs/2312.10997.

[55] P. Xu, W. Ping, X. Wu, L. McAfee, C. Zhu, Z. Liu, S. Subramanian, E. Bakhturina, M. Shoeybi, B. Catanzaro, Retrieval meets long context large language models, 2024, arXiv:2310.03025. URL: https://arxiv.org/abs/2310.03025.

[56] W. Kryściński, N.S. Keskar, B. McCann, C. Xiong, R. Socher, Neural text summarization: A critical evaluation, 2019, arXiv preprint arXiv:1908.08960.

[57] D. Li, B. Jiang, L. Huang, A. Beigi, C. Zhao, Z. Tan, A. Bhattacharjee, Y. Jiang, C. Chen, T. Wu, et al., From generation to judgment: Opportunities and challenges of LLM-as-a-judge, 2024, arXiv preprint arXiv:2411.16594.

[58] L. Parcalabescu, A. Frank, On measuring faithfulness or self-consistency of natural language explanations, 2023, arXiv preprint arXiv:2311.07466.

[59] S. Kim, J. Shin, Y. Cho, J. Jang, S. Longpre, H. Lee, S. Yun, S. Shin, S. Kim, J. Thorne, et al., Prometheus: Inducing fine-grained evaluation capability in language models, in: The Twelfth International Conference on Learning Representations, 2023.

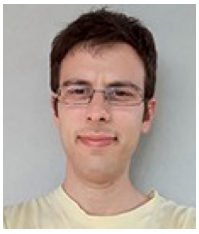
[60] Z. Li, B. Peng, P. He, X. Yan, Evaluating the instruction-following robustness of large language models to prompt injection, in: Y. Al-Onaizan, M. Bansal, Y.-N. Chen (Eds.), Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing, Association for Computational Linguistics, Miami, Florida, USA, 2024, pp. 557–568.



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